

# MBA in International Business

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FINAL REPORT

## ***AI Washing or Credible Disclosure? Patterns and Clusters in Corporate AI Disclosures***

By

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I declare that this assignment is my own work, which all sources of reference are acknowledged in full, and it has not been submitted for any other course.

Signature:



How AI disclosure is measured

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## EXECUTIVE SUMMARY

Corporate AI disclosure expanded rapidly after generative AI became commercially salient: between 2021 and 2025, the share of S&P 500 firms making an AI-related statement rose from 36% to 92%. This thesis asks whether the content of that disclosure distinguishes talk from disclosed substance, and whether that distinction is economically useful.

Three findings organize the evidence. First, firms' AI disclosures fall into recurrent profiles, but only the profile reporting the most concrete activity is consistently economically distinct after sector-year adjustment. Second, specificity and disclosed activity add information beyond disclosure volume and financial fundamentals; promotional tone and the within-disclosure gap between talk and disclosed substance do not. Third, the same firm-year presents a similar activity mix in statutory filings and earnings calls, but calls attach substantially more concrete evidence and promotion. That venue-associated gap widened with the 2023 boom and shows no detectable closure after the SEC's 2024 scrutiny.

The practical implication is not that all promotional AI language is empty. Rather, readers should distinguish volume from specificity, disclosed activity, and cross-venue consistency. The thesis supplies a scalable measurement pipeline and a credibility framework for issuers, investors, and regulators; its claims are associative, not causal.

## 1. INTRODUCTION

### 1.1 BUSINESS PROBLEM

The release of ChatGPT in November 2022 turned artificial intelligence from a specialist topic into a mainstream corporate disclosure issue. In this corpus, between 2021 and 2025 the share of S&P 500 firms making at least one AI-related statement in their SEC filings rose from 36% to 92%, and the narrative density of AI disclosure (AI statements per 1,000 words of corporate narrative text) grew nearly seven-fold (from 0.016 to 0.111 per 1,000 words). The single largest jump occurs in the 2024 filing season, when fiscal-year-2023 10-Ks become the first annual reports written with a full year of generative AI behind them: the share of 10-Ks containing any AI statement jumps from 39% to 72% in one season. Independent work also documents a sharp post-ChatGPT increase in 10-K AI disclosure (Anantharaman, Parker, and Rozario 2026, 1–2).

This is, at the same time, an investor problem, an issuer problem, and a regulatory problem:

- **Investors** need to know whether a firm's AI narrative reflects real capability, real deployment, and real economic exposure, or whether it is positioning attached to a hot topic. Studies of filings distinguish actionable or implemented AI from speculative claims and relate that distinction to valuation or later outcomes (Basnet et al. 2025, 1–2; Song et al. 2026, 1–2).
- **Issuers** need to know how much disclosure is enough, and in which venue, without inviting enforcement risk.
- **Regulators**, the SEC prominently among them, have already acted. In February 2024, Chair Gensler warned about "AI washing" (Gensler 2024, 4–5), and the SEC's March 2024 actions

against Delphia and Global Predictions alleged false and misleading statements about AI use under the Advisers Act and Marketing Rule (U.S. Securities and Exchange Commission 2024, 1–2). Those actions make the enforcement risk concrete; they do not establish that every promotional AI statement is unlawful.

The SEC’s April 2025 comment letter to Welltower illustrates why venue comparisons are useful. The staff requested that future filings discuss its data-science platform and the status of AI integration, after noting both the 10-K’s statement that it was integrating generative-AI tools and the earnings-call and press-release descriptions of the platform as “industry-leading” and a source of competitive advantage (U.S. Securities and Exchange Commission 2025, 1). The letter neither finds any statement false nor defines a general cross-venue test; it motivates, rather than validates, the thesis’s cross-venue measurement.

Mention counts, a common measure in finance and disclosure research, tracing back to Loughran and McDonald’s (2011, 36) word lists, cannot answer the semantic questions posed here. A count of how many times “artificial intelligence” appears in a 10-K conflates a risk-factor boilerplate sentence with a specific claim about a named, deployed product generating measured revenue. It cannot:

- separate promotional register from governance register;
- tell whether a claim describes something the firm has done versus something it might do; or
- compare what a firm says in a call against what it commits to in a filing.

Answering the business problem (is the disclosure substantive, and does it matter economically) requires moving past counting mentions and building a measurement pipeline that extracts what is actually claimed: who is the subject, what type of AI, in what tense, in what register, and, one layer deeper, what concrete activity, with what evidence, is behind the claim.

## 1.2 RESEARCH QUESTIONS

The overarching question this thesis answers is: **can the content of corporate AI disclosure distinguish talk from disclosed substance, and does that distinction carry useful economic information?**

Four linked questions organize the empirical chapters:

- **RQ1: Evolution and composition.** How has corporate AI disclosure evolved, and what activities do firms report?
- **RQ2: Heterogeneity.** What recurrent disclosure profiles distinguish firms, and how do those profiles relate to firm economics?
- **RQ3: Credibility and decoupling.** Where do gaps emerge between disclosure and disclosed substance, and between statutory filings and earnings calls?
- **RQ4: Economic relevance.** Does AI disclosure carry economic information and relate to market outcomes?

## 1.3 LITERATURE ON CORPORATE AI DISCLOSURE AND AI WASHING

### 1.3.1 THE EMERGING EMPIRICAL FRONTIER

The explosion of commercial interest in artificial intelligence following the release of ChatGPT triggered a rapid empirical literature. Studies examine AI language in annual filings (Basnet et al. 2025, 1; Song et al. 2026, 1–2; Anantharaman, Parker, and Rozario 2026, 1–2) and firm-level generative-AI exposure constructed from occupational tasks and employment shares (Eisfeldt,

Schubert, and Zhang 2023, 9). This work builds on the broader textual-analysis tradition in finance (Li 2008, 1–3; Loughran and McDonald 2011, 35–36; Gentzkow, Kelly, and Taddy 2019, 1–2).

Simultaneously, a regulatory and market concern has arisen: “AI washing,” the use of AI claims that are false, misleading, or unsupported by observable implementation. Gensler used the term in his warning and the SEC used it in the Delphia and Global Predictions release (Gensler 2024, 4–5; U.S. Securities and Exchange Commission 2024, 1–2). Borrowing conceptual scaffolding from greenwashing and corporate environmental decoupling (Delmas and Burbano 2011, 64–68; Lyon and Montgomery 2015, 1–3; Marquis, Toffel, and Zhou 2016, 483–85), recent work contrasts speculative AI narratives with observable implementation or actionable disclosures (Song et al. 2026, 1–2; Basnet et al. 2025, 1–2).

### 1.3.2 FOUR GAPS IN THE CURRENT LITERATURE

Despite rapid progress, the existing literature on corporate AI disclosure and AI washing has four gaps:

1. **Keyword retrieval alone treats mentions too coarsely.** Some AI-disclosure studies begin with keyword retrieval from annual reports or 10-Ks (Anantharaman, Parker, and Rozario 2026, 1–2). Others improve on it by classifying mentions as actionable, speculative, or irrelevant (Basnet et al. 2025, 1; Song et al. 2026, 1–2). Keyword retrieval by itself cannot tell an empty risk-factor sentence (“new AI tools developed by competitors could disrupt our market position”) apart from a specific claim about a named, deployed product with a measured result (“we deployed an internal predictive model that cut supply-chain logistics costs by 14%”).
2. **The cited studies do not provide the same activity inventory.** Basnet et al. classify filing narratives into actionable, speculative, and irrelevant mentions; Song et al. distinguish planned from implemented AI using text and patents; and Eisfeldt et al. construct firm-level generative-AI exposure from task and occupational-employment data (Basnet et al. 2025, 1; Song et al. 2026, 1–2; Eisfeldt, Schubert, and Zhang 2023, 9). These designs answer different questions from a structured inventory of the business function, technology, sourcing, maturity, and evidence attached to each disclosed activity.
3. **The cited AI studies use annual filings rather than a matched call–filing design.** Earnings calls are a voluntary disclosure channel (Matsumoto, Pronk, and Roelofsen 2011, 1383–84), and research on calls documents managers’ incentives and discretion in that channel (Frankel, Johnson, and Skinner 1999, 133–34). Comparing the same firm-period in calls and filings therefore adds a venue dimension absent from the annual-filing designs cited above. The Welltower letter is a concrete example in which the SEC staff referenced both types of communication (U.S. Securities and Exchange Commission 2025, 1).
4. **No complete external census validates firm-level AI adoption.** Greenwashing research can compare claims with environmental performance indicators (Delmas and Burbano 2011, 68–69; Marquis, Toffel, and Zhou 2016, 486–87). AI research can use partial observables such as patents (Song et al. 2026, 1–2), but these do not constitute a complete census of deployment across business functions. The thesis therefore measures disclosed activity and its grounding, not actual adoption.

### 1.3.3 WHAT THIS THESIS DOES ABOUT IT

This thesis addresses these four gaps in four measurement and design steps:

1. **From keyword counts to structured claims.** Instead of counting words or computing hard-to-interpret text embeddings, the pipeline filters millions of paragraphs with an interpretable

classifier, then has an LLM judge extract one structured claim per statement: who is speaking, what type of AI, in what tense (realized, planned, expected, hypothetical), about what business domain, using which concept from a fixed vocabulary, how specific, how promotional, and with what sentence-level evidence.

2. **An inventory of what firms say they do, not just a number for a regression.** Instead of reducing disclosure text to a single score, this thesis builds a descriptive inventory of 27,883 concrete activity claims (action, object, and business function) across 476 S&P 500 firms, recording the stage, audience, evidence, and technology source (in-house vs. third-party) behind each one. This is an inventory of what firms *say* they do, not an audited record of what they operationally do.
3. **Comparing the same firm across venues.** Firm  $\times$  fiscal-year fixed effects compare the call and the filing while holding the firm and period fixed. This design measures a venue-associated difference; it does not isolate the causal effect of legal liability, because audience, format, and topic selection also vary with venue.
4. **A single AI-washing index, covering every firm-year that discloses AI.** Rather than an arbitrary tercile cut or a claim-by-claim ratio restricted to a highly promotional subsample, the thesis defines AI washing as the percentile-rank gap between disclosure intensity and concrete, grounded disclosed activity. The measure is defined for every firm-year that mentions AI and is examined alongside the cross-venue comparison as a distinct form of decoupling.

## 1.4 THEORETICAL FOUNDATIONS

Three ideas from economics, accounting, and organizational sociology motivate the empirical designs that follow.

**Signalling and cheap talk.** Spence's (1973, 363–64) signalling model makes signal cost central to credibility; voluntary-disclosure theory likewise models the costs of releasing information (Dye 1985, 123–24). Crawford and Sobel (1982, 1431–32) analyze strategic information transmission when sender and receiver interests diverge. These models do not prove that a named model, quantified gain, or commercial partner is costly or true. They motivate the empirical distinction: such details are potentially more verifiable than generic promotional language, while tone itself can be associated with litigation risk (Rogers, Van Buskirk, and Zechman 2011, 2155–56).

**Institutional pressure to look current.** Organizational sociology (Meyer and Rowan 1977, 340–41; DiMaggio and Powell 1983, 147–48; Suchman 1995, 571–74) argues that under technological uncertainty, firms adopt the language and symbols expected of them to stay legitimate, whether or not that language reflects any change in what they actually do. This is a plausible reason why AI disclosure spread from 36% to 92% of the S&P 500 in four years, including in sectors where AI has limited operational relevance.

**Talk that doesn't match conduct.** Meyer and Rowan (1977, 340–41) distinguish formal, ceremonial structure from ongoing activity; Westphal and Zajac (1994, 367–69) and Crilly, Zollo, and Hansen (2012, 1429–31) examine forms of policy–practice decoupling. Crilly et al. contrast intentional decoupling (“faking it”) with implementation friction (“muddling through”). In corporate disclosure, related concerns appear in work on impression management and tone (Merkl-Davies and Brennan 2007, 116–18; Huang, Teoh, and Zhang 2014, 1083–86). These theories motivate the thesis's disclosed-activity measures; they do not turn a disclosure–activity gap into proof of actual nonimplementation.

**Four terms used throughout (disclosure intensity, disclosed behaviour, activity grounding, AI washing):**

- **AI disclosure intensity:** how much a firm says about AI, measured from the semantic-frame pass as AI frames per 1,000 words of a firm's own filings that year. It is distinct from *promotional posture*, the share of a firm's AI statements written in promotional, strategic, or aspirational language.
- **Disclosed behaviour (disclosed activity volume):** measured from a separate LLM extraction pass over concrete, action-level claims (Chapter 2's disclosed-activity inventory), kept deliberately apart from the frame-level measures used to build disclosure intensity, so the two remain distinct instruments rather than restatements of the same content, though both still read the same underlying text, so any relationship between them is descriptive, not independently confirmed.
- **Activity grounding:** a firm-year score for how concretely disclosed activities are specified (named business function, deployed/scaled stage, named product, quantified outcome, named provider).
- **AI washing (the index):** the percentile-rank gap between AI disclosure intensity and disclosed substance — disclosed activity volume weighted by activity grounding — defined only for firm-years that disclose AI at all. It is one within-disclosure form of decoupling; the related cross-venue gap compares calls and filings.

#### Propositions mapped to empirical research questions.

- *RQ1 (Evolution and composition):* Documents the growth and changing composition of AI disclosure, then inventories the activities firms report.
- *RQ2 (Heterogeneity):* Tests whether disclosure posture organizes into stable archetypes and whether those profiles correspond to economically distinct firms after sector-year adjustment.
- *RQ3 (Credibility and decoupling):* Measures the within-disclosure percentile gap and the within-firm, within-year call-versus-filing gap; the SEC exercise is a secondary identification check within this question.
- *RQ4 (Economic relevance):* Evaluates whether disclosure content carries incremental economic information via post-call beta regressions, isolating historical disclosure and substantive activity from call-specific surprises.

## 1.5 CONTRIBUTION

The thesis makes three contributions.

**1. Measurement.** A scalable five-layer pipeline separates AI disclosure *volume*, *semantic content*, and *disclosed activity* across SEC documents and earnings-call transcripts.

**2. Empirical design.** The analysis separates the evolution and composition of disclosure, recurrent disclosure profiles, incremental economic information, and credibility gaps rather than treating them as a single mention-count exercise.

**3. Credibility framework.** A within-disclosure disclosure–substance gap and a within-firm, cross-venue comparison distinguish two forms of decoupling without treating either as direct evidence of actual operational conduct.

**Managerial contribution.** For issuers, the thesis converts the measurement into a concrete checklist for credible disclosure (named activity, clear stage, quantified outcome, provider transparency, venue consistency) and quantifies a venue gap regulators are already scrutinizing. For investors, it identifies which textual dimensions carry incremental information (specificity and disclosed activity,

not promotional tone) and shows that cross-venue consistency between the call and the filing is itself a usable credibility signal.

## 1.6 THESIS ROADMAP

- **Chapter 2** documents the corpus, the measurement pipeline, and its human validation.
- **Chapter 3** establishes corporate AI disclosure strategies: documenting the diffusion of AI into corporate language and showing how firms differentiate along seven posture dimensions into stable, recurrent disclosure archetypes.
- **Chapter 4** maps disclosure strategy to disclosed activity: inventorying what corporate AI actually consists of across 27,883 activities, testing how communication posture corresponds to operational claims, revealing the paradox of the GenAI boom, and demonstrating external economic distinctiveness.
- **Chapter 5** tests the credibility of corporate AI disclosure across three progressively demanding empirical dimensions: internal substance, cross-venue consistency, and regulatory scrutiny response.
- **Chapter 6** evaluates the economic and market relevance of AI disclosure via earnings-call beta regressions, testing whether disclosure distinctions carry incremental information or reflect pre-existing firm characteristics.
- **Chapter 7** draws together the issuer, investor, and theoretical implications without repeating the empirical results.
- **Chapter 8** concludes.

Table 1: Theory to empirical test.

| Theory                                                                                                                                            | What is measured                                                           | Hypothesis / RQ | Empirical test                                                                                           | Chapter |
|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-----------------|----------------------------------------------------------------------------------------------------------|---------|
| Institutional diffusion / legitimacy (DiMaggio and Powell 1983, 147–48; Suchman 1995, 571–74); Recurrent disclosure archetypes (Hennig 2007, 1–2) | Share of firms disclosing, seven posture dimensions, K-means clusters      | RQ1 & RQ2       | Diffusion 2021–2025, posture distribution, K-means, bootstrap stability, sector composition              | 3       |
| Grounded operational conduct (Meyer and Rowan 1977, 340–41); Economic coherence (Spence 1973, 363–64)                                             | 27,883 activity claims, substance ladder, sector-year fundamental profiles | RQ2             | Activity map, archetype × activity profile, 100% role mix, GenAI boom evolution, sector-year regressions | 4       |
| Credibility and observable decoupling (Meyer and Rowan 1977, 340–41; Crilly, Zollo, and Hansen 2012, 1429–31)                                     | Substance scaling ( <i>W</i> ); cross-venue evidence persistence           | RQ3             | Percentile ranks; firm × fiscal-year fixed effects                                                       | 5       |
| Enforcement credibility (Gensler 2024, 4–5; U.S. Securities and Exchange Commission 2024, 1–2; Callaway, Goodman-Bacon, and Sant’Anna 2024, 1–2)  | Pre/post SEC scrutiny                                                      | RQ3             | Event study with continuous exposure, DiD                                                                | 5       |
| Signalling / cheap talk (Spence 1973, 363–64; Crawford and Sobel 1982, 1431–32)                                                                   | Historical disclosure and substance vs. call-specific surprises            | RQ4             | Post-call beta regressions, expanding histories, ANCOVA                                                  | 6       |

## 2. DATA AND MEASUREMENT

### 2.1 CORPUS AND SOURCES

The U.S. corpus covers 10-K, 10-Q, 20-F, DEF 14A, and 8-K filings for the S&P 500 universe plus earnings-call transcripts, from 2021 through a partial 2026. The document counts and deduplication totals below are calculated directly from the current processed corpus at render time; the U.S. instrument carries embeddings, prefilter scores, and semantic frames, and is the basis for every result in this thesis.

The current processed corpus contains **58,352 documents** and **9,783,273 paragraph instances**. The length-and-content filter retains 8,934,740 scorable instances, which deduplicate to 5,123,286 unique texts.

Table 2: Current processed U.S. corpus by form.

| Form          | Documents |
|---------------|-----------|
| 10-K          | 2,898     |
| 10-Q          | 8,330     |
| 20-F          | 32        |
| DEF 14A       | 2,883     |
| 8-K           | 35,880    |
| Earnings call | 8,329     |
| Total         | 58,352    |

The measurement pipeline moves through analytical screening stages: from the raw corpus through content filtering and deduplication, a machine-learning prefilter, semantic-frame extraction, and disclosed-activity extraction. The unit of analysis changes at each stage (document, deduplicated text, individual claim, concrete activity) because each stage needs a different unit to do its job. Conceptually, this multi-stage inference architecture instantiates the classical Knowledge Discovery in Databases (KDD) process (Fayyad, Piatetsky-Shapiro, and Smyth 1996, 40–41)—progressing through selection, preprocessing, transformation, data mining, and pattern evaluation—and aligns with the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework (Shearer 2000, 15–19), mapping regulatory and corporate understanding into disciplined, reproducible data engineering and model evaluation stages. [Figure 1](#) illustrates this end-to-end data acquisition, inference, and multi-source fusion architecture.

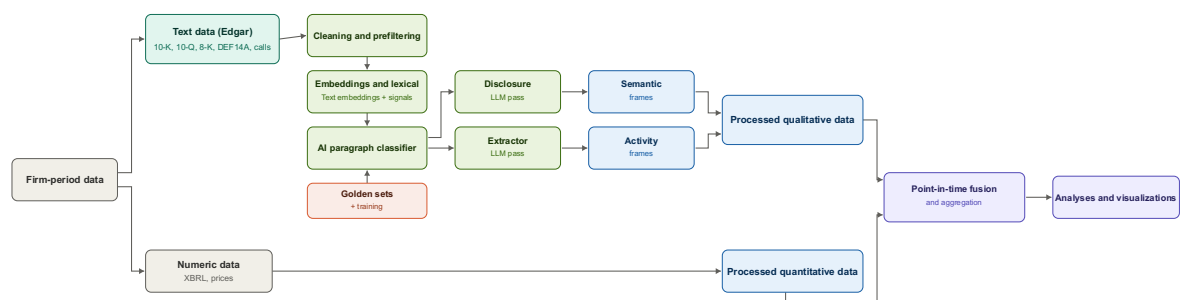


Figure 1: End-to-end data acquisition, inference, and multi-source fusion pipeline.

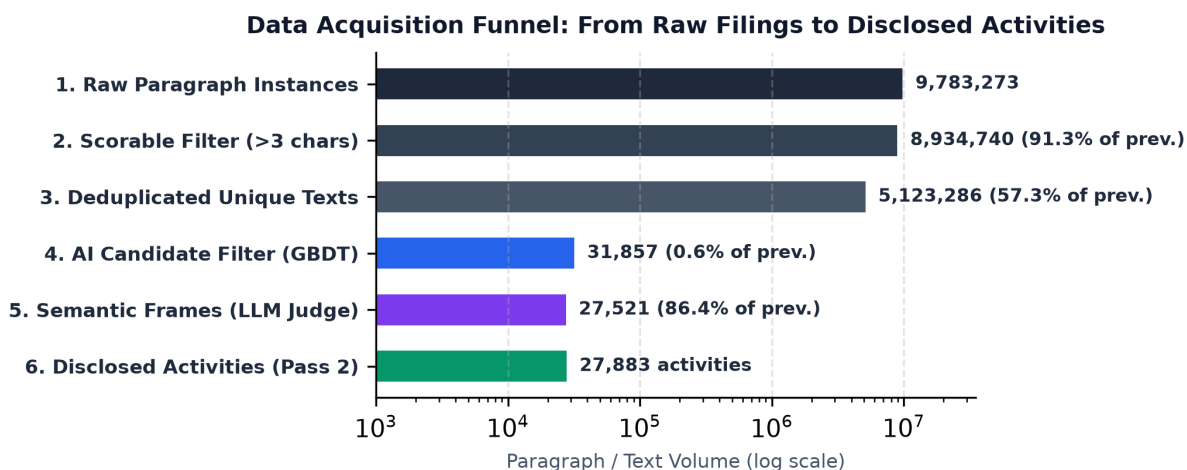


Figure 2: The Data Acquisition Funnel: progression of corpus volume across analytical screening stages (log scale).

Table 3: Compact summary of the inference funnel.

| Stage                    | Unit of Observation       | Operational Filter                             | Retained Volume | Share of Raw |
|--------------------------|---------------------------|------------------------------------------------|-----------------|--------------|
| 1. Raw Corpus            | Paragraph instances       | Raw segmentation across 58,352 filings & calls | 9,783,273       | 100.00%      |
| 2. Scorable Filter       | Scorable instances        | Length (>3 chars) & alphanumeric content       | 8,934,740       | 91.33%       |
| 3. Deduplication         | Unique scorable texts     | BLAKE2b content hash                           | 5,123,286       | 52.37%       |
| 4. Machine Learning Gate | Candidate paragraphs      | Supervised GBDT classifier (tau=0.17)          | 31,857          | 0.33%        |
| 5. Semantic Frames       | Texts with >=1 frame      | Zero-shot LLM judge (qwen3.7-flash)            | 27,521          | 0.28%        |
| 6. Disclosed Activities  | Behavioral action triples | Firm-conditioned LLM extraction (pass 2)       | 27,883          | 0.29%        |

*Notes: High-level overview of the six-stage inference funnel. Granular form-by-form decompositions (10-K, 10-Q, DEF 14A, 8-K, Earnings Calls, and 20-F) and intermediate model acceptance diagnostics are reported in [Table 26](#) (Appendix C).*

## 2.2 UNITS OF ANALYSIS

The pipeline moves across seven nested units, each used where it is the natural denominator:

- the unique text (a deduplicated paragraph, the unit of scoring and classification);
- the semantic frame (one claim extracted from a text; a text can yield zero, one, or several frames);
- the disclosed activity (one action-object-function triple extracted from a behavioural frame);
- the document (the unit of intensity, since a text's frames are joined back to every document instance in which it appears);
- the firm-year (the unit of the economic analysis);
- the pooled firm (the unit of the cross-sectional segmentation and washing score); and
- the firm × fiscal-year × venue cell (the unit of the cross-venue comparison).

A firm-quarter unit is used only for the SEC and DeepSeek event studies, where calendar-quarter alignment is required to define a clean pre/post cut.

## 2.3 CLEANING, PREFILTERING, AND AI PARAGRAPH CLASSIFIER

Before any LLM reads a paragraph, the text processing stack cleans, deduplicates, and classifies every candidate paragraph. Every scorable paragraph is hashed on its normalized text, which collapses 8.93 million paragraph instances into 5.12 million unique texts. Everything downstream runs once per unique hash rather than once per document, and results are joined back afterward to every filing that repeats that paragraph. This keeps recycled boilerplate and legal disclaimers from being scored repeatedly across documents.

Text embeddings and lexical signals are computed for each unique paragraph. A gradient-boosted classifier evaluates 39 signals across four groups:

- keywords (a hand-built list of AI terms and roughly 60 named entities);
- meaning (cosine similarity to known AI language using dense sentence embeddings);
- shape (length, capitalization, position in the document); and
- sentence-level cues.

It is trained on a 9,900-text golden set labeled by two independent LLM judges ([Table 6](#)) and scores 0.96 on the standard F1 metric for 10-K filings, meaning it rarely misses a real AI mention and rarely flags something irrelevant, on the document type it was tuned for. Precision is recovered downstream in the LLM passes, not here.

On a held-out sample never used in training, precision (the share of flagged texts that are actually about AI) stays at 0.98 for 10-K/10-Q, drops to 0.73 for proxy statements and 8-Ks, and drops further to 0.59 for earnings calls. Recall (the share of true AI mentions the filter catches) stays above 0.92 everywhere, so the filter almost never misses something real; it just lets through more noise on some document types than others.

In total, it flags 31,857 texts as AI-relevant, 0.6% of everything scored. Every downstream number in this thesis counts a document with zero flagged texts as a documented zero, not as a missing observation: a firm that never mentions AI is data, not a gap in the sample.

## 2.4 DISCLOSURE LLM PASS: SEMANTIC FRAMES

Every text the prefilter flags is then read once by an LLM. Prior computational social-science work finds that LLM labels can be comparable to human labels in specific, evaluated tasks (Gilardi, Alizadeh, and Kubli 2023, 1–2); that evidence does not remove the need for the held-out evaluation reported here. The pass pulls out zero, one, or several separate claims about AI, a “frame.” Each frame records eight things about the claim:

- who it’s about (the firm itself, a customer, a competitor, a supplier);
- what kind of AI (generative vs. predictive/ML: generative went from roughly half of mentions in 2021 to 93% by 2026);
- its tense (already happening, planned, expected, or hypothetical: 72% describe something already happening);
- its business domain;
- which concept from a fixed list it matches (adoption, capability, risk, or governance);
- how specific it is;
- whether it reads as promotional; and

- what evidence backs it.

This step also acts as a second, more careful filter: 14% of what the cheap prefilter flagged turns out, on a closer read, to have zero real AI claims and gets dropped. In total, 31,857 texts are read this way, yielding 48,787 distinct claims.

Table 4: Semantic frame schema: the eight fields recorded per claim, with observed value distributions.

| Field       | Description                                                                    | Observed values                                                                      |
|-------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| subject     | Who the claim is about                                                         | firm (84%); competitors_or_industry (8%); customers (7%); suppliers_or_partners (1%) |
| ai_type     | Generative vs. predictive/ML                                                   | unspecified (56%); generative (38%); predictive_ml (6%)                              |
| temporal    | Tense: realized, planned, expected, hypothetical                               | realized (71%); planned (15%); expected (10%); hypothetical (5%)                     |
| domain      | Internal, customer-facing, or unspecified                                      | customer_facing (44%); internal (42%); unspecified (13%)                             |
| concepts    | Fixed vocabulary of adoption/capability/risk/governance concepts (multi-label) | reported descriptively by frame type                                                 |
| specificity | 5 sub-flags: process, product, vendor, metric, date/timeline named             | combined into a firm-level specificity measure                                       |
| rhetoric    | Promotional or strategic-importance framing (boolean)                          | 14.9% promotional (corpus mean)                                                      |
| evidence    | Sentence indices supporting the claim                                          | always non-empty by construction                                                     |

## 2.5 EXTRACTOR LLM PASS: ACTIVITY FRAMES

Whenever a claim describes the firm itself doing something concrete (deploying, investing, hiring, measuring a result), it goes through a second pass. This time the model is told which company it's reading, so it can tell "our own product" apart from "a tool we bought." For each such claim it records:

- the action (deploy, develop, partner, invest in infrastructure, hire, pilot, scale, measure a result, govern, restrict, and a few others);
- what the action targets (in free text, later grouped into families) and who benefits from it (employees, customers, internal process, partners);
- how mature it is (exploring, piloting, deployed, scaled);
- whose technology it is (the firm's own, a third party's, co-developed, acquired, open-source);
- who else gets named (a partner, a vendor, a competitor, a customer); and
- how solid the evidence is (a named product, a number, a named vendor, or just generic language).

Every one of these records is tied to an actual sentence; nothing is inferred without a supporting quote. This pass produces 27,883 distinct activities across 476 of the 510 firms.

Table 5: Disclosed activity schema: the fields recorded per activity, with observed value distributions.

| Field  | Description          | Observed values                                                                  |
|--------|----------------------|----------------------------------------------------------------------------------|
| action | What the firm did    | deploy (63%); invest_infrastructure (8%); scale (7%); pilot_or_explore (4%); ... |
| stage  | Operational maturity | deployed (58%); scaled (14%); piloting (10%); exploring (9%); ...                |

| Field             | Description                                                                                               | Observed values                                                                 |
|-------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| ai_source         | Whose technology it is                                                                                    | own (73%); unspecified (16%); third_party (8%); acquired (1%); ...              |
| target            | Who benefits                                                                                              | customers (48%); internal_process (26%); unspecified (14%); employees (9%); ... |
| evidence_strength | How solid the evidence is                                                                                 | generic (54%); named_product_or_process (29%); metric (13%); vendor (4%)        |
| object            | Free text, later grouped into 21 object families                                                          | reported as an action × object typology                                         |
| named_entities    | Own brand, external provider/model, partner, acquired company, customer, distribution channel, competitor | 17,310 entity mentions total                                                    |

## 2.6 POINT-IN-TIME FUSION AND AGGREGATION

The measurement pipeline distinguishes between two complementary concepts of disclosure volume:

1. **Diffusion:** the share of filings (or of firms) with at least one AI statement. Bounded between 0 and 1, and unaffected by how many separate claims a document makes.
2. **Cumulative frame density:** the total number of AI statements per 1,000 words of scorable narrative text: how much AI content there is, not just whether it's present. Every "per 1,000" intensity figure in this thesis, including in the clustering, regression, and event-study analyses, uses words as the denominator rather than paragraphs, so a single unit of measurement holds throughout.

The two move together closely (a firm with more AI content is almost always also a firm that discloses at all), so it makes little difference which one a ranking or a regression uses. On average, a paragraph that discusses AI makes 1.44 separate statements, rising from 1.43 in 2021 to 1.49 in 2025 as filings got denser.

To go from a single document to a firm-year, each document's rate is combined with the others using the median, not an average weighted by length. The reason is practical: a firm-year can include one 10-K and several 10-Qs and 8-Ks of very different sizes, and the median stops one unusually long or short document from dominating the number. In practice this choice barely matters, because the 10-Q almost never contains an AI statement, so a firm-year's AI content usually comes from a single filing (its 10-K) anyway, and the median gives the same answer as a length-weighted average to the ninth decimal place across the analytical firm-year panel (checked directly, not assumed).

For concepts describing behaviour (deployment, risk, governance, etc.), the default measure is a log-transformed count rather than a share of a firm's total AI statements: the count version turned out to be much more reliable when the same firm was checked against itself twice (a reliability score of 0.845 vs. 0.528 for shares).

Firm-level rates are also pulled slightly toward the corpus average when a firm has very few AI statements to base a rate on, so a firm with two mentions doesn't get treated as confidently as one with two hundred; a firm with zero mentions gets no rate at all and goes into its own "no AI" category instead.

Activity grounding is simpler: the average of five yes/no questions per activity:

- is the function named;
- is it deployed or scaled;
- is the product named;

- is there a number attached; and
- is the provider named.

## 2.7 GOLDEN SETS AND MODEL TRAINING

Every automated stage in the measurement pipeline is benchmarked against held-out ground truth and inter-annotator evaluation datasets. [Table 6](#) details the evaluation datasets, their sample design, and empirical performance metrics.

Table 6: Labeled evaluation datasets, validation benchmarks, and inter-judge agreement.

| Benchmark Dataset                 | Forms        | Sample Size   | Annotated By                 | Metric / Empirical Performance                       | Status      |
|-----------------------------------|--------------|---------------|------------------------------|------------------------------------------------------|-------------|
| <b>Golden Set (Training)</b>      | 10-K, 10-Q   | 9,900 texts   | Qwen 3.7 / Gemini 3.8        | F1 = 0.96 (Prec. 0.98, Rec. 0.94)                    | Verified    |
| <b>Inter-Judge Agreement</b>      | 10-K, 10-Q   | 6,038 pairs   | Gemini 3.8 vs. Qwen 3.7      | Cohen's $\kappa$ = 0.81–0.87 across dimensions       | Verified    |
| <b>Out-of-Domain Proxy</b>        | DEF 14A, 8-K | 1,500 texts   | Qwen 3.7 Flash               | Prec. 0.73, Rec. 0.92, F1 = 0.82 (14A), 0.65 (8-K)   | Verified    |
| <b>Out-of-Domain Calls</b>        | Transcripts  | 800 texts     | Qwen 3.7 Flash               | Prec. 0.59, Rec. 0.98, F1 = 0.73 (Strong Prec. 0.85) | Verified    |
| <b>Lexical Anchors &amp; Gate</b> | All forms    | 27 anchors    | Expert curated               | 39 signals, BGE-M3 cosine anchors                    | Verified    |
| <b>Activity Extraction</b>        | Mixed sample | 30 paragraphs | Iterative manual inspection  | Entity role & source prompt engineering              | Verified    |
| <b>Human Ground Truth</b>         | All forms    | 720 texts     | Human coders (ui-validator/) | Human-to-LLM agreement (434 frames, 267 activities)  | In progress |

*Notes:* All operational labels supporting the empirical chapters derive from zero-shot LLM extraction. The measured inter-judge agreement ( $\kappa$  = 0.81–0.87) establishes strong reproducibility across distinct foundation model architectures (Google Gemini vs. Alibaba Qwen).

## 2.8 PROCESSED QUANTITATIVE DATA: FINANCIAL AND MARKET VARIABLES

Firm-year financial and market variables are drawn from XBRL and price data, aligned by actual fiscal period end rather than filing date, with documented fallback chains for concept coverage (capex 91%, SG&A 81%, net margin 99%, revenue 99%) and a [340, 380]-day gap rule for validating year-over-year changes. The variable set used throughout Chapters 3–6 comprises market capitalization, R&D-to-sales, gross margin, operating margin, asset turnover, price-to-sales, next-fiscal-year revenue growth, market beta, and idiosyncratic volatility (the annualized standard deviation of one-factor market-model residuals over the 252 trading days preceding the filing). R&D-to-sales has only 46% XBRL coverage and is used as an outcome, not a general control: it identifies firms that separately report comparable current R&D rather than treating non-reporting as zero. Appendix D gives the beta specification, its estimation-window diagnostics, the treatment of delisted firms, and the R&D coverage assessment. All financial ratios are winsorized at the 1st and 99th percentiles.

## 3. CORPORATE AI DISCLOSURE STRATEGIES

### 3.1 AI BECOMES UBIQUITOUS

Between 2021 and 2025, artificial intelligence transformed from a specialist topic into a standard element of corporate disclosure. Across the S&P 500, the share of firms making at least one AI statement in their SEC filings rose from 35.7% in 2021 to 92.1% in 2025 (Figure 3). Over the same period, narrative intensity—the volume of AI semantic frames per 1,000 words of corporate text—expanded nearly seven-fold, climbing from 0.016 to 0.111 frames per 1,000 words.

The sharpest inflection occurs around the 2023–2024 filing season, when fiscal-year-2023 10-Ks became the first annual reports written with a full year of generative AI commercialization behind them: the share of 10-Ks containing any AI content surged from 39% to 72% in a single season. A balanced cohort of the 475 firms present continuously across 2021–2026 reproduces this identical trajectory, confirming that this reflects market-wide adoption rather than survivorship or index rebalancing.

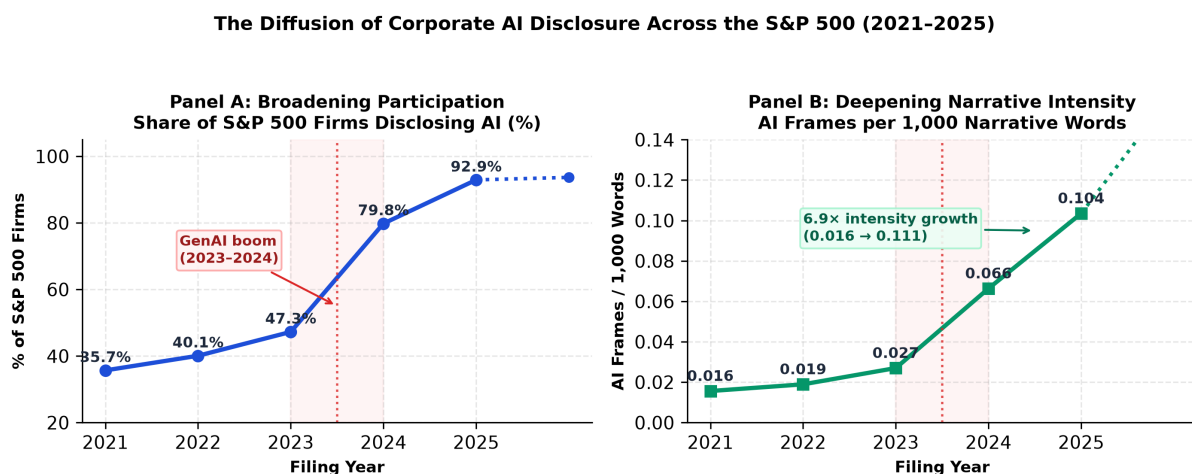


Figure 3: The diffusion of corporate AI disclosure across the S&P 500 (2021–2025): broadening participation in percentage of disclosing firms (Panel A) and deepening narrative intensity in AI frames per 1,000 words (Panel B), with the generative AI commercialization inflection marked.

Once almost every large firm talks about AI, mere presence ceases to be informative. When participation reaches 92.1%, the binary question of *whether* a firm discloses AI no longer differentiates corporate behavior. Instead, differentiation shifts to *how* firms communicate: the framing, emphasis, risk perception, and specificity that define their corporate disclosure posture.

### 3.2 THE DIMENSIONS OF AI DISCLOSURE STRATEGY

To capture how firms position their AI disclosures without reducing text to crude sentiment scores or unanchored keyword tallies, the measurement pipeline extracts six conceptually distinct posture dimensions directly from gold\_ai\_frames. Each dimension represents a simple, named average of binary indicators shrunk empirically toward the corpus mean in proportion to frame support (Chapter 2’s empirical-Bayes aggregation rule):

- **Promotional posture:** the average of promotional tone and strategic-importance rhetoric.
- **Risk orientation:** the share of statements carrying a risk concept (uncertainty, dependency, or downside exposure).

- **Governance orientation:** the share of statements carrying a governance concept (oversight bodies, ethics policies, compliance, or steering committees).
- **Temporal posture:** the share of statements describing realized or actively implemented actions, rather than planned, expected, or hypothetical future initiatives.
- **AI positioning:** the share framed as customer-facing products or client solutions rather than internal process enhancements.
- **Specificity:** the average of five binary concreteness flags (named business process, named product, named vendor, quantified metric, or explicit timeline).

A seventh dimension, **disclosure intensity** (a percentile rank of AI frames per 1,000 words), distinguishes firms that employ similar language registers at markedly different volumes of communication.

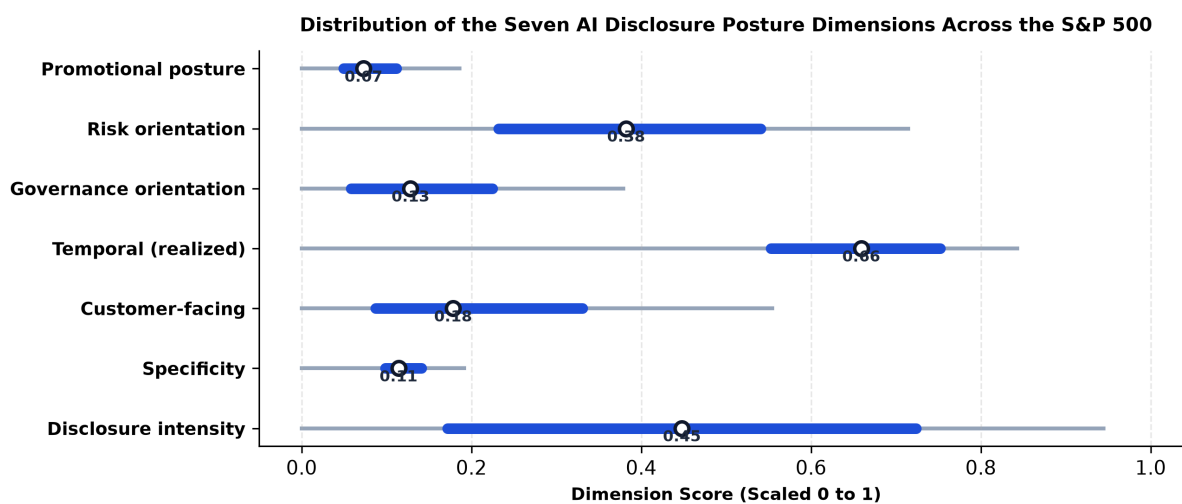


Figure 4: Distribution of the seven AI disclosure posture dimensions across the S&P 500 (white circle = median; blue bar = interquartile range; grey whisker = 5th–95th percentile).

Table 7: Operational definitions of the seven AI disclosure strategy dimensions.

| Dimension              | Operational Definition                                                        | Range | Managerial Meaning                                                           |
|------------------------|-------------------------------------------------------------------------------|-------|------------------------------------------------------------------------------|
| Promotional posture    | Average of promotional tone and strategic-importance framing                  | [0,1] | Extent to which AI is communicated as a transformative strategic asset       |
| Risk orientation       | Share of statements containing risk, uncertainty, or dependency concepts      | [0,1] | Degree to which AI is treated as an operational vulnerability or hazard      |
| Governance orientation | Share of statements containing oversight, ethics, or compliance policies      | [0,1] | Formal institutionalization of model oversight and responsible AI guardrails |
| Temporal posture       | Share of statements describing realized rather than planned actions           | [0,1] | Operational maturity: active deployment vs. speculative roadmap              |
| AI positioning         | Share of statements framed toward clients and external products               | [0,1] | Commercial orientation: revenue generation vs. internal cost-cutting         |
| Specificity            | Mean of 5 binary grounding flags (process, product, vendor, metric, timeline) | [0,1] | Falsifiability and informational granularity of corporate statements         |

| Dimension            | Operational Definition                                          | Range | Managerial Meaning                                                             |
|----------------------|-----------------------------------------------------------------|-------|--------------------------------------------------------------------------------|
| Disclosure intensity | Percentile rank of AI semantic frames per 1,000 narrative words | [0,1] | Overall volume of narrative communication dedicated to artificial intelligence |

Examining the correlation structure of these seven dimensions reveals two critical structural properties of corporate communication (full correlation matrix reported in [Table 22](#) in Appendix C):

1. **Governance framing is largely orthogonal to other disclosure dimensions:** Governance orientation exhibits minimal correlation with promotional posture ( $r = 0.00$ ), risk framing ( $r = -0.12$ ), and disclosure intensity ( $r = -0.20$ ). A firm's commitment to governance disclosure operates as an independent communication layer rather than a proxy for general verbosity or risk aversion.
2. **Risk orientation directly opposes concrete execution:** Risk framing correlates negatively with realized temporal posture ( $r = -0.48$ ), specificity ( $r = -0.58$ ), customer-facing positioning ( $r = -0.48$ ), and intensity ( $r = -0.37$ ). Firms that frame AI primarily as an operational hazard speak less about it, describe fewer concrete artifacts, and rarely report client-facing deployments.

### 3.3 RECURRENT DISCLOSURE ARCHETYPES

Firms with fewer than five AI frames across their pooled 2021–2025 SEC filings are assigned the rule-based label **No AI** ( $n = 49$ ). This filter prevents assigning an artificial strategic posture to firms that barely mention the technology.

For all remaining firms ( $n = 461$ ), K-means clustering is applied to the seven standardized posture dimensions. The clustering partitions corporate communication into three distinct, interpretable archetypes ([Figure 5](#)):

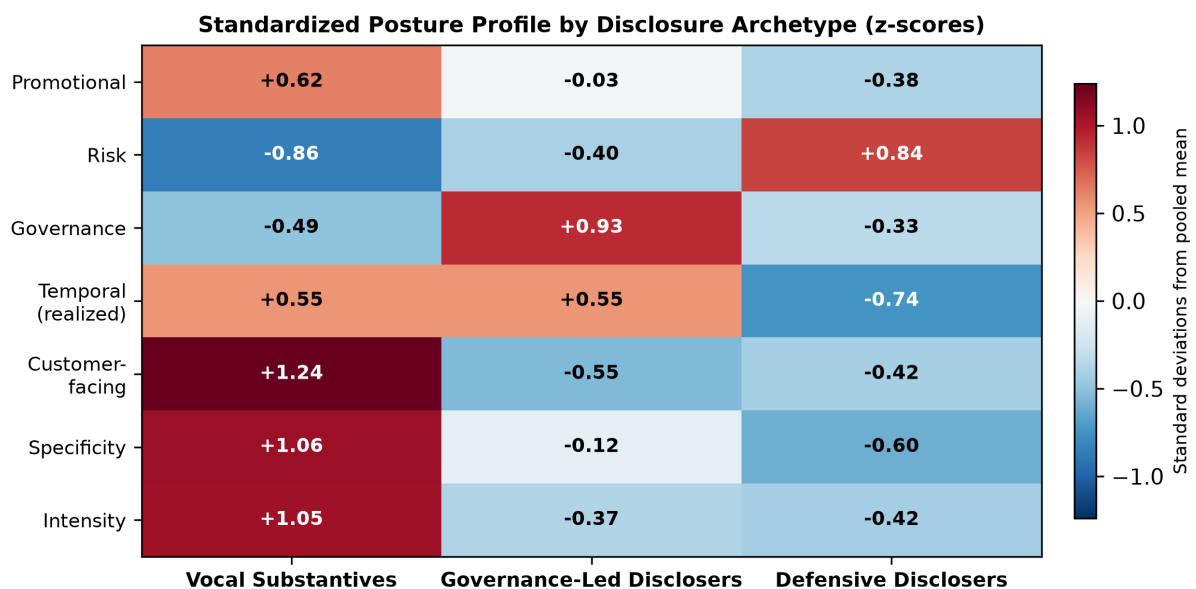


Figure 5: Disclosure archetypes at a glance: standardized (z-score) mean posture profile of the three K-means archetypes.

Table 8: AI corporate disclosure archetypes at a glance. Year-over-year archetype persistence is 69.3%.

| Archetype                        | Firms (n) | Share (%) | Core Posture Characteristics                                                                                 | Stability |
|----------------------------------|-----------|-----------|--------------------------------------------------------------------------------------------------------------|-----------|
| <b>Vocal Substantives</b>        | 127       | 24.9%     | Highest promotional (+0.65 SD), customer-facing (+0.84 SD), specificity (+0.92 SD), and intensity (+1.12 SD) | 0.75      |
| <b>Governance-Led Disclosers</b> | 138       | 27.1%     | Above-average governance (+1.28 SD); moderate promotional (-0.15 SD) and specificity (-0.12 SD)              | 0.81      |
| <b>Defensive Disclosers</b>      | 196       | 38.4%     | Highest risk (+0.85 SD); lowest realized (-0.72 SD), customer-facing (-0.68 SD), and specificity (-0.68 SD)  | 0.66      |
| <b>No AI</b>                     | 49        | 9.6%      | Fewer than 5 pooled AI frames across 2021–2025 SEC filings (rule-based filter)                               | 1.00      |

The three empirical archetypes display sharply contrasting communication postures:

- **Vocal Substantives (127 firms, 24.9% of sample):** The technology frontrunners. Defined by high promotional posture (+0.65 SD), the highest customer-facing positioning (+0.84 SD), the highest specificity (+0.92 SD), and the highest narrative intensity (+1.12 SD). They report concrete, deployed applications and promote them assertively.
- **Governance-Led Disclosers (138 firms, 27.1% of sample):** The compliance and oversight posture. Characterized exclusively by above-average governance orientation (+1.28 SD), with moderate promotional tone (-0.15 SD), moderate specificity (-0.12 SD), and below-average risk orientation (-0.38 SD).
- **Defensive Disclosers (196 firms, 38.4% of sample):** The risk-dominated majority. The largest group across the S&P 500, defined by high risk orientation (+0.85 SD) and below-average realized execution (-0.72 SD), customer orientation (-0.68 SD), specificity (-0.68 SD), and intensity (-0.78 SD). These firms treat AI predominantly as an external uncertainty to be hedged in mandatory risk disclosures.

Within the panel, firms exhibit substantial stability: year-over-year archetype persistence is **69.3%**, confirming that disclosure archetypes reflect durable communication policies rather than transitory single-year filing noise.

### 3.4 ARE THE ARCHETYPES REAL AND STABLE?

To verify that these three clusters reflect genuine structure rather than algorithmically forced partitions, we evaluate cluster stability through non-parametric bootstrap resampling. For each firm, we resample its own constituent AI frames with replacement 25 times, re-estimating the full pipeline—recomputing posture dimensions and re-running K-means—and computing the mean Jaccard similarity between the resampled clusters and the original assignments.

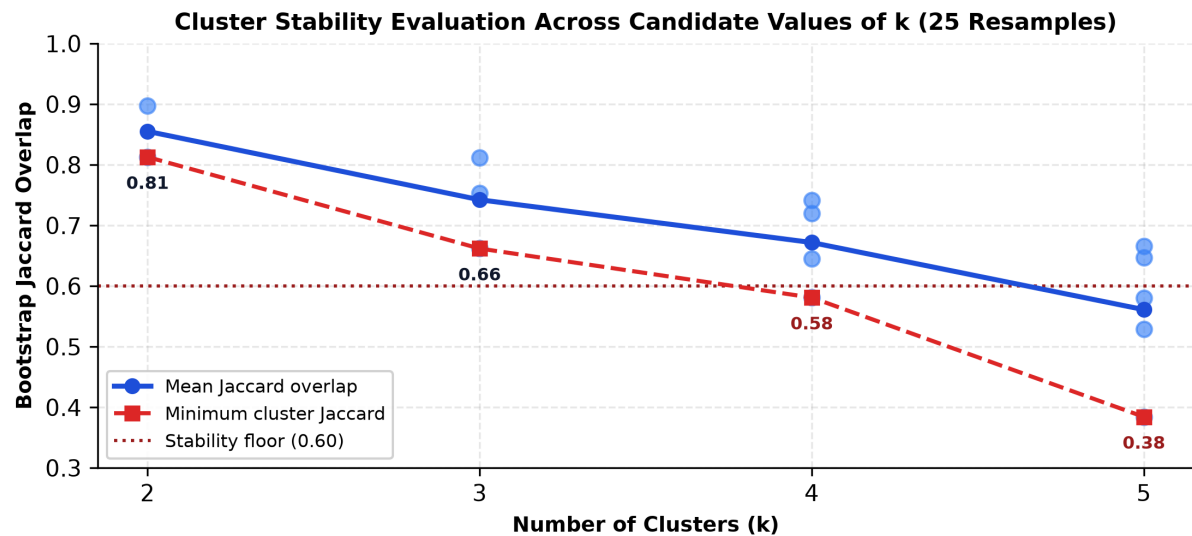


Figure 6: Choosing  $k$  by bootstrap stability: cluster stability evaluation across candidate values of  $k$  (25 resamples of each firm's own frames). Only  $k = 3$  keeps every cluster above the 0.60 Jaccard floor.

As shown in [Figure 6](#),  $k = 3$  is the largest number of clusters where every individual cluster clears the 0.60 mean-Jaccard stability floor (Vocal Substantives: 0.75, Defensive Disclosers: 0.66, Governance-Led: 0.81). At  $k = 4$ , the worst-recovered cluster collapses to 0.58, and at  $k = 5$  to 0.38. Choosing  $k = 3$  (yielding four groups when combined with the deterministic No AI category) maximizes granular differentiation while guaranteeing empirical replicability.

**PCA diagnostic.** A principal components analysis on the seven standardized dimensions confirms that this space cannot be reduced to a single score without severe information loss. The first two components account for 63.3% of the variance (PC1: 43.2%, PC2: 20.2%), while PC3 adds an additional 12.1% (full diagnostic in [Table 23](#), Appendix C). Because variance is distributed across multiple orthogonal factors, preserving the seven interpretable managerial dimensions provides far clearer strategic guidance than collapsing communication into abstract latent factors.

### 3.5 WHO ARE THESE FIRMS?

Do these communication archetypes map onto specific sectors, or do firms within the same industry adopt divergent disclosure postures?

Comparing observed sector counts against sample baseline expectations reveals strong sectoral sorting ([Figure 7](#)):

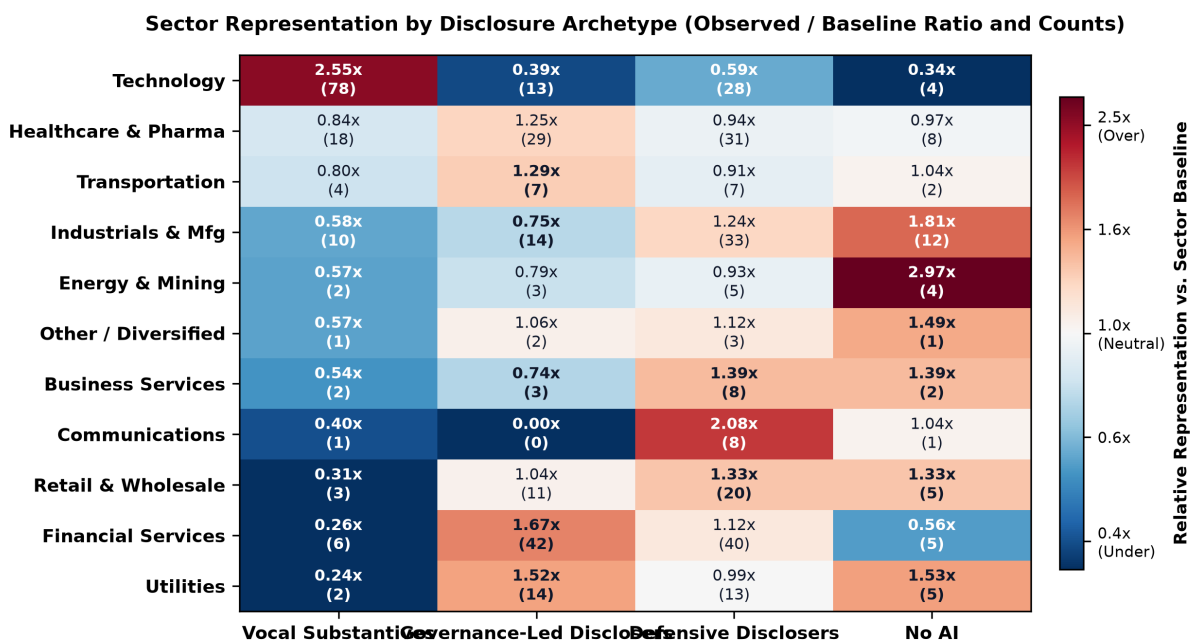


Figure 7: Sector representation by disclosure archetype: ratio of archetype share to sample baseline (observed / expected ratio, with firm counts in parentheses). Technology concentrates heavily in Vocal Substantives; Financials and Utilities in Governance-Led; Communications and Consumer in Defensive.

- **Technology** is heavily concentrated in **Vocal Substantives** (2.55× overrepresented relative to baseline; 78 of 127 firms). Technology providers possess both the technical capabilities and the market incentive to aggressively publicize customer-facing AI products.
- **Financial Services** (1.67× overrepresented; 42 firms), **Utilities** (1.52×), and **Healthcare & Pharma** (1.25×) dominate **Governance-Led Disclosers**. These highly regulated sectors operate under stringent compliance mandates, prompting them to prioritize ethical guardrails, model risk management, and oversight frameworks.
- **Communications** (2.08×), **Business Services** (1.39×), **Retail & Wholesale** (1.33×), and **Industrials & Manufacturing** (1.24×) are heavily overrepresented in **Defensive Disclosers**. In these sectors, AI presents competitive threats or internal process adjustments rather than core revenue products.
- **Energy & Mining** (2.97× overrepresented) and **Industrials** (1.81×) account for the vast majority of **No AI** firms, where digital transformation remains secondary to physical operations.

Table 9: Representative corporations across AI disclosure archetypes.

| Archetype                        | Centroid-Closest Firms                                         | Major Prominent Corporations                                          | Dominant Strategic Sector              |
|----------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------|
| <b>Vocal Substantives</b>        | GE HealthCare (GEHC), HPE (HPE), Verisk (VRSK), Trimble (TRMB) | NVIDIA (NVDA), Microsoft (MSFT), Adobe (ADBE), Alphabet (GOOGL)       | Technology & Electronics (61.4%)       |
| <b>Governance-Led Disclosers</b> | Biogen (BIIB), Avery Dennison (AVY), 3M (MMM), Corteva (CTVA)  | JPMorgan Chase (JPM), Apple (AAPL), UnitedHealth (UNH), Walmart (WMT) | Financials (30.4%), Healthcare (21.0%) |

| Archetype                   | Centroid-Closest Firms                                                | Major Prominent Corporations                                                      | Dominant Strategic Sector             |
|-----------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------|
| <b>Defensive Disclosers</b> | Constellation Brands (STZ), Bio-Rad (BIO), Constellation Energy (CEG) | Nike (NKE), Goldman Sachs (GS), Caterpillar (CAT), ExxonMobil (XOM)               | Industrials, Retail, Communications   |
| <b>No AI</b>                | Boundary threshold                                                    | Berkshire Hathaway (BRK.B), Costco (COST), Marathon Petroleum (MPC), Valero (VLO) | Energy & Mining, Physical Industrials |

As illustrated in [Table 9](#), examining individual firms demonstrates the face validity of the taxonomy: technology leaders like Microsoft and NVIDIA anchor Vocal Substantives; heavily regulated giants like JPMorgan Chase, Apple, and UnitedHealth anchor Governance-Led Disclosers; mature consumer and industrial brands like Nike, Goldman Sachs, and Caterpillar fall into Defensive Disclosers; and resource-focused companies like Berkshire Hathaway and Valero Energy omit AI disclosure entirely.

### 3.5.1 FROM COMMUNICATION POSTURE TO OPERATIONAL REALITY

This chapter has established how S&P 500 corporations communicate about artificial intelligence once the topic becomes ubiquitous: they separate into three robust, stable disclosure archetypes.

Yet an essential question remains: **are these communication postures mere corporate rhetoric, or do they correspond to fundamentally different operational commitments?** To determine whether a firm's disclosure posture reflects real-world differences in how AI is utilized, Chapter 4 transitions from communication strategy to the concrete activities firms claim to perform.

## 4. FROM DISCLOSURE STRATEGY TO DISCLOSED ACTIVITY

### 4.1 WHAT CORPORATE AI ACTUALLY CONSISTS OF

To evaluate whether disclosure archetypes correspond to distinct operational profiles, we open the black box of corporate disclosure using the second-stage extraction pipeline: the **disclosed-activity inventory**. Across 476 S&P 500 firms (93.3% of disclosing firms), this extraction identifies 27,883 unique, concrete activity claims, recording the action, object, business domain, operational stage, evidence strength, and technology origin behind each claim.

To make these activities managerially interpretable, we categorize them into six empirical business roles:

- **Operate (Internal operations & workflows):** Optimizing supply chains, automating business processes, operational forecasting, logistics, and back-office financial administration.
- **Serve (Sales, marketing & client service):** Customer service chatbots, conversational client interfaces, recommendation engines, and targeted marketing.
- **Sell (Customer products & digital services):** Embedding AI capabilities, generative features, and predictive analytics directly into client-facing products and software solutions.
- **Enable (Employee productivity & software development):** Developer copilots, automated code generation, enterprise knowledge search, and internal employee tooling.
- **Control (Risk, cybersecurity & governance):** Real-time fraud detection, cybersecurity anomaly detection, compliance monitoring, and model risk oversight.
- **Build (Compute, data platforms & foundational models):** Enterprise data platforms, scalable cloud infrastructure, and training proprietary foundation models.

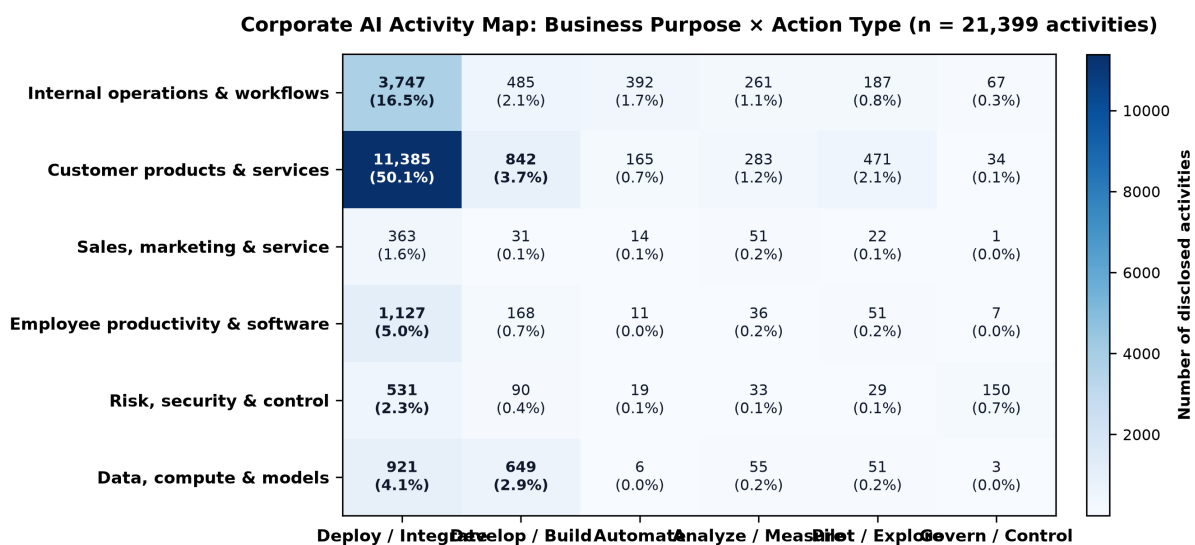


Figure 8: Corporate AI activity map: heatmap of disclosed activities across business purpose and operational action type (n = 21,399 categorized activities across 476 firms).

The cross-tabulation in [Figure 8](#) establishes a foundational empirical reality: **corporate AI is overwhelmingly an application-layer phenomenon**. S&P 500 firms describe deploying, integrating, and embedding commercial AI tools into existing customer products and operational workflows. In contrast, developing foundation models or building core computing infrastructure accounts for a negligible fraction of total enterprise activity.

## 4.2 DISCLOSURE STRATEGY MEETS DISCLOSED ACTIVITY

The central empirical test of this chapter connects communication posture with disclosed conduct. Crucially, **the disclosed-activity inventory was never an input to clustering**: the archetypes were identified strictly from frame-level communication posture.

Cross-tabulating the archetypes against disclosed activity characteristics demonstrates that communication posture maps directly onto operational disclosure ([Figure 9](#) and [Table 10](#)):

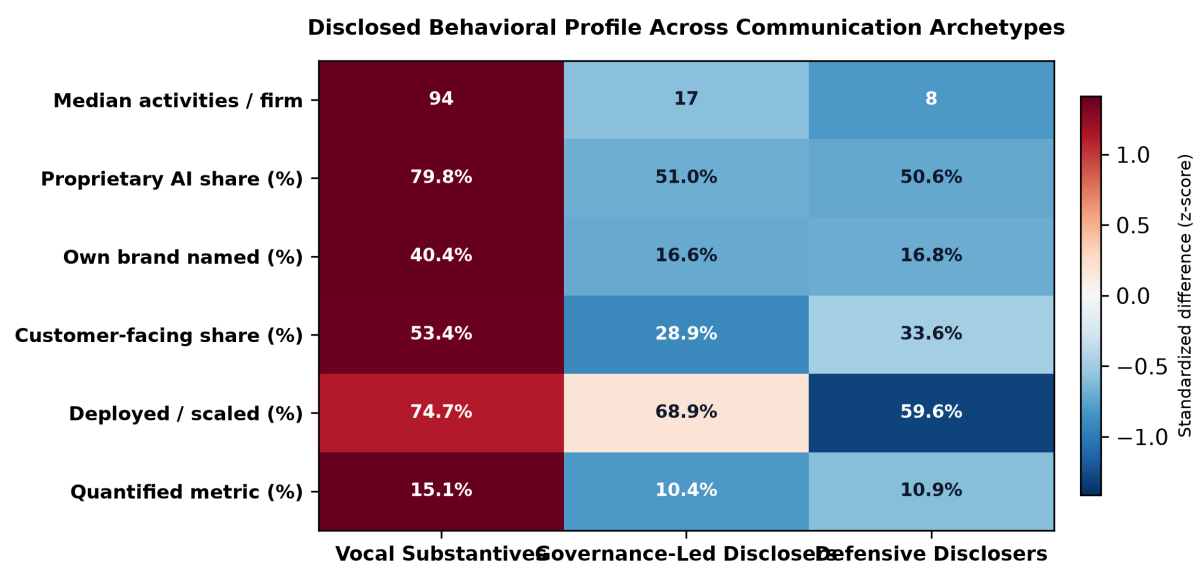


Figure 9: Disclosed behavioral profile across communication archetypes. Cell values report raw behavioral rates; shading indicates standardized intensity across archetypes.

Table 10: Disclosed behavioral profile by disclosure archetype, from the disclosed-activity inventory (never an input to the posture clustering).

| Behavioral feature                            | Vocal Substantives<br>(n=127) | Governance-Led Disclosers<br>(n=138) | Defensive Disclosers<br>(n=196) | No AI<br>(n=49) |
|-----------------------------------------------|-------------------------------|--------------------------------------|---------------------------------|-----------------|
| Disclosed activities per firm (median)        | 94                            | 17                                   | 8                               | 0               |
| Customer-facing activity share (%)            | 53.4%                         | 28.9%                                | 33.6%                           | 31.8%           |
| Technology source: proprietary/own (%)        | 79.8%                         | 51.0%                                | 50.6%                           | 45.5%           |
| Activities naming an own product/brand (%)    | 40.4%                         | 16.6%                                | 16.8%                           | 18.0%           |
| Activities reporting a quantified outcome (%) | 15.1%                         | 10.4%                                | 10.9%                           | 13.7%           |
| Activities at stage deployed/scaled (%)       | 74.7%                         | 68.9%                                | 59.6%                           | 59.7%           |

*Notes:* Distributions computed from the 27,883-activity inventory across 510 S&P 500 firms, joined post hoc to the posture-based archetype assignment. “No AI” firms with positive activity counts reflect conference-call statements by firms whose formal SEC filings omit AI entirely. Full 14-variable behavioral profile reported in [Table 25](#) (Appendix C).

Across all six behavioral dimensions, the archetypes separate cleanly:

- **Vocal Substantives** report a median of **94.0 activities per firm**—more than five times the volume of Governance-Led Disclosers (17.0) and eleven times Defensive Disclosers (8.5). Furthermore, 80.0% of their activities claim proprietary technology, 40.7% name a specific corporate brand or product, and 74.7% are deployed in active production.
- **Governance-Led Disclosers** display an intermediate profile: a median of 17.0 activities per firm, with 51.7% claiming proprietary AI and 61.6% reaching deployment.
- **Defensive Disclosers** report only 8.5 activities per firm, with the lowest deployed share (59.5%) and only 18.1% naming a specific product.

**Controlling for disclosure volume.** Because both semantic frames and activities are extracted from the same corporate text, could this behavioral gap be a trivial artifact of verbosity? To rule out this shared-source confound, we estimate a negative binomial regression of disclosed activity counts on disclosure archetypes while explicitly conditioning on total AI disclosure volume ( $\log(1 + \text{AI frames})$ ):

$$\log E[\text{Activities}_i] = \alpha + \beta_1 \text{VocalSubstantives}_i + \beta_2 \text{GovernanceLed}_i + \gamma \log(1 + \text{Frames}_i)$$

Table 11: Disclosed activity count by archetype, before and after conditioning on total AI disclosure volume (log frame count). Reference category: Defensive Disclosers.

| Archetype vs. Defensive Disclosers | Raw ratio | p      | Volume-adjusted ratio | p      |
|------------------------------------|-----------|--------|-----------------------|--------|
| Governance-Led Disclosers          | 1.64x     | <0.001 | 1.38x                 | <0.001 |
| Vocal Substantives                 | 10.86x    | <0.001 | 2.33x                 | <0.001 |
| $\log(1 + \text{AI frames})$       | -         | -      | +1.00 (coef.)         | <0.001 |

Total AI disclosure volume explains most of the raw gap (log frame count alone reaches  $R^2 = 0.74$  in the adjusted model), exactly as the shared-source-text concern predicts. But a real difference survives conditioning on it: at the same total disclosure volume, Vocal Substantives still report **2.44 times as many activities** as Defensive Disclosers ( $p < 0.001$ ) and Governance-Led Disclosers report **1.41 times as many** ( $p < 0.001$ ). Communication posture is not merely verbosity; it reflects fundamentally different intensities of disclosed operational execution.

### 4.3 WHAT ARE FIRMS ACTUALLY DOING DIFFERENTLY?

The behavioral distinction between archetypes extends beyond activity counts to the substantive business purpose of their AI initiatives ([Figure 10](#)):

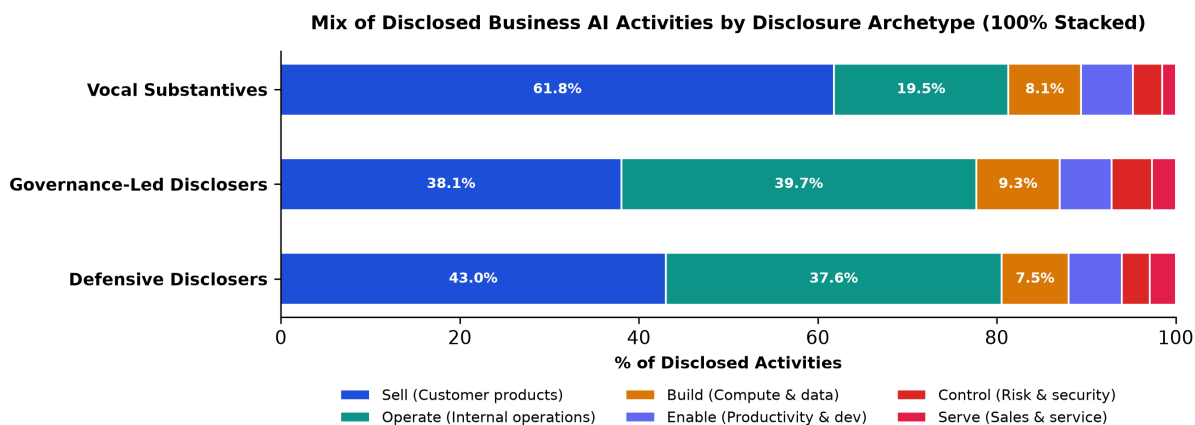


Figure 10: Mix of disclosed business AI activities by disclosure archetype (100% stacked horizontal bars). Vocal Substantives dedicate over 60% of activities to client products (Sell); Governance-Led prioritize internal operations (Operate) and governance control (Control).

The 100% stacked distribution in [Figure 10](#) reveals stark strategic divergences in where AI enters the firm:

1. **Vocal Substantives focus on revenue-generating products:** Nearly two-thirds of their disclosed activities (**61.8%**) belong to **Sell** (embedding AI directly into customer products and services), while internal operations account for only 19.5%.

2. **Governance-Led Disclosers emphasize operational efficiency and control:** Their activity mix is led by **Operate** (39.7%) and **Sell** (38.1%), but they devote the highest share among all groups to **Control** (4.5% in risk, fraud detection, and model governance).
3. **Defensive Disclosers mirror back-office automation:** Their activities split between **Sell** (43.0%) and **Operate** (37.6%), with heightened focus on **Serve** (2.9% in customer service bots), reflecting utility and efficiency use cases rather than technological leadership.

## 4.4 NOT ALL ACTIVITY IS EQUALLY SUBSTANTIVE

While firms disclose thousands of AI activities, a critical empirical question is whether these claims represent verifiable operational reality or vague corporate aspirations. To evaluate claim substance, we benchmark all 27,883 activities against an empirical **ladder of disclosed substance** that imposes increasing informational standards:

Activity Claim → Specific Business Function → Deployed Stage → Named Product/Process  
→ Quantified Outcome Metric

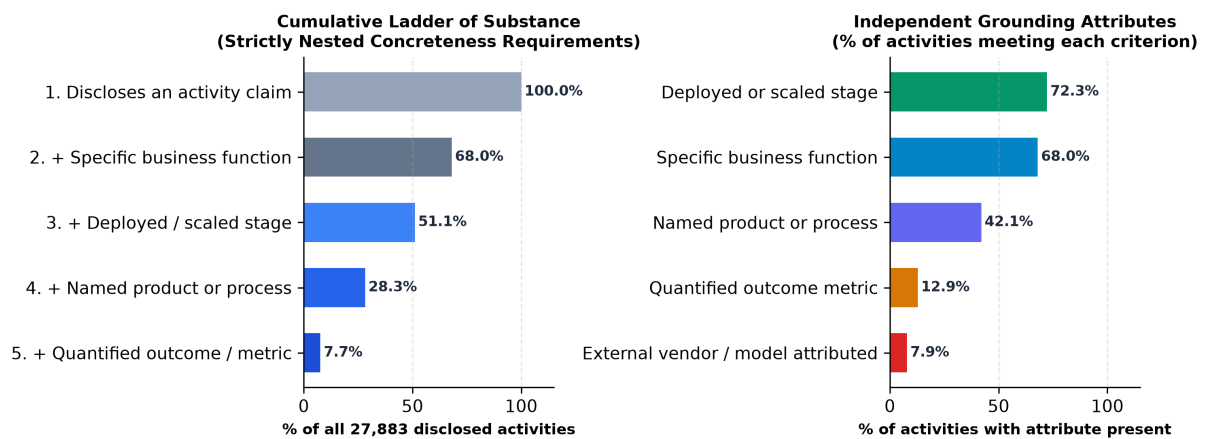


Figure 11: The ladder of disclosed substance: cumulative nested progression of evidence requirements (left) and independent grounding attribute frequencies (right) across 27,883 disclosed activities.

As shown in [Figure 11](#), while 68.0% of activities specify a business function and 51.1% are deployed in production, the share that names a tangible product or process drops to 28.3%. Furthermore, only 7.7% of all disclosed activities satisfy the full ladder by coupling an operational deployment to a quantified performance metric. Most disclosed AI activities remain difficult to externally evaluate because they are not accompanied by quantified outcomes.

## 4.5 THE PARADOX OF THE GENAI BOOM

Tracking these grounding dimensions over time reveals a central empirical paradox: **did the post-2023 generative AI boom produce more grounded corporate disclosure?**

The evidence indicates the exact opposite: **activity volume surged, but marginal concreteness deteriorated** ([Figure 12](#)).

### The Paradox of the GenAI Boom: Disclosed Activity Expanded Faster than Observable Grounding

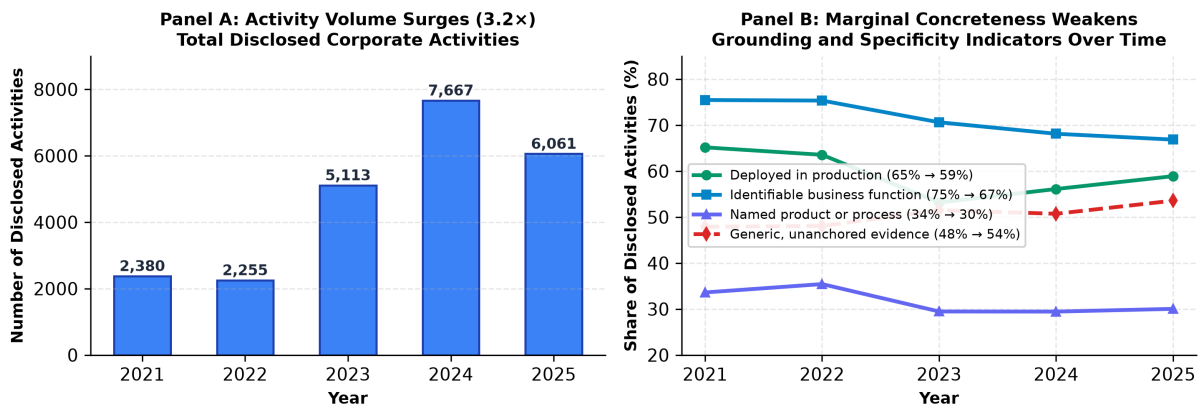


Figure 12: The paradox of the GenAI boom: activity volume surged 3.2-fold (Panel A), while marginal concreteness and grounding indicators deteriorated (Panel B), 2021–2025.

Crucially, this finding does not claim that real-world firm capabilities deteriorated. Rather, it establishes an informational phenomenon: **the marginal corporate AI activity disclosed following ChatGPT was less concrete, less mature, and more speculative than the disclosures preceding it.**

## 4.6 WHO OWNS THE AI?

A fundamental question for investors and regulators evaluating corporate AI claims is technological provenance: are public firms genuine technology creators, or commercial consumers of foundation models developed by third-party hyperscalers?

Comparing named entity roles across the 27,883 activity statements exposes a marked provenance asymmetry (Figure 13):

- **35.0%** of all activity statements explicitly attach a proprietary corporate brand or in-house product name to the AI deployment.
- **Only 4.4%** explicitly cite an external technology vendor or foundation model (and only 2.2% name a commercial partner).
- This occurs despite **48.2%** of all S&P 500 firms disclosing at least one external technology provider, model family, or hyperscaler somewhere within their broader disclosures.

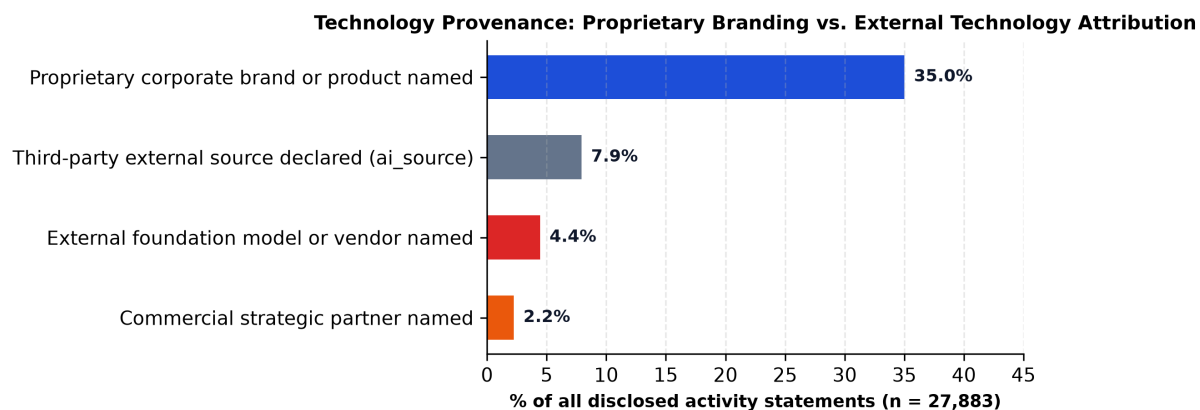


Figure 13: Technology provenance: proprietary branding vs. external vendor attribution across 27,883 corporate AI activities.

When external vendors are identified, the ecosystem is highly concentrated around four hyperscalers: OpenAI / Microsoft Ecosystem (cited by 17.3% of firms), Alphabet / Google (8.2%), NVIDIA (5.3%), and Amazon Web Services (4.9%). The complete vendor taxonomy ([Table 29](#)) and named entity distribution ([Table 28](#)) are reported in Appendix C.

Corporate disclosures disproportionately foreground proprietary brands while identifying the external technology stack far less frequently.

## 4.7 EXTERNAL VALIDATION: DO DISCLOSURE AND ACTIVITY PROFILES CORRESPOND TO DIFFERENT FIRMS?

Unlike the activity cross-tabulation above, the financial and market variables used here never passed through the LLM extraction pipeline at all; they come from XBRL filings and market prices, an entirely separate data source from the text the archetypes are built on. This is genuine external evidence, but it answers a narrower question than “are the archetypes real”: it asks whether the archetypes correspond to economically distinct *firms*, not whether the posture dimensions themselves are valid. An archetype with no economic footprint would still be a real, reproducible communication strategy; it would just mean that strategy is not one that shows up in fundamentals, which is itself an informative finding, not a failure of the clustering.

The economic-distinctiveness analysis pools all firm-years with filings from 2021 to 2025 (2,475 firm-years, 508 firms; archetype counts: 1,425 No AI, 450 Vocal Substantives, 350 Defensive Disclosers, 250 Governance-Led Disclosers) and tests whether the archetypes correspond to economically distinct firms, both crudely and after sector-by-year adjustment ( $X = \alpha[\text{sector} \times \text{year}] + \gamma \text{Archetype} + \epsilon$ , referenced against No AI, firm-clustered standard errors).

Crude medians already put Vocal Substantives ahead of the other two archetypes on the dimensions that matter most for a technology-intensive segment: R&D/sales 12.4% vs. 5.7–6.2% for the other archetypes, gross margin 61.1% vs. 41.5–44.8%, beta 1.09 vs. 0.79–0.81. The residualized panel, within the same sector and year, confirms each archetype separately:

- **Vocal Substantives** is the most consistently distinct archetype: +0.44 log points of market cap ( $p < 0.001$ ), +5.6 percentage points of R&D-to-sales ( $p < 0.001$ ), +11.0 points of gross margin ( $p < 0.001$ ), +0.10 of beta ( $p = 0.008$ ), and +0.97 of price-to-sales ( $p = 0.028$ ), significant on five of the nine dimensions tested.

- **Governance-Led Disclosers** separate from No AI mainly on scale and earnings quality rather than technology intensity: a market-cap premium (+0.33 log points,  $p = 0.002$ ) and a ROIC-minus-WACC premium (+0.065,  $p = 0.007$ ), but *lower* price-to-sales (−0.86,  $p = 0.023$ ) and no R&D or margin premium. This is consistent with the archetype being defined by governance-heavy disclosure that correlates with firm size and regulatory exposure (larger, more heavily regulated firms disclose more governance content) rather than with distinctive AI economics.
- **Defensive Disclosers** barely separate from No AI at all: a small R&D premium (+3.6 points,  $p = 0.010$ ) is the only dimension reaching significance out of nine tested.

No archetype shows a robust advantage in every dimension tested. The story that survives sector-year adjustment is that the archetype most concretely grounded in disclosed conduct (Vocal Substantives, [Table 10](#)) is also the one most consistently distinct on firm fundamentals, not that talking about AI, or emphasizing governance, is by itself.

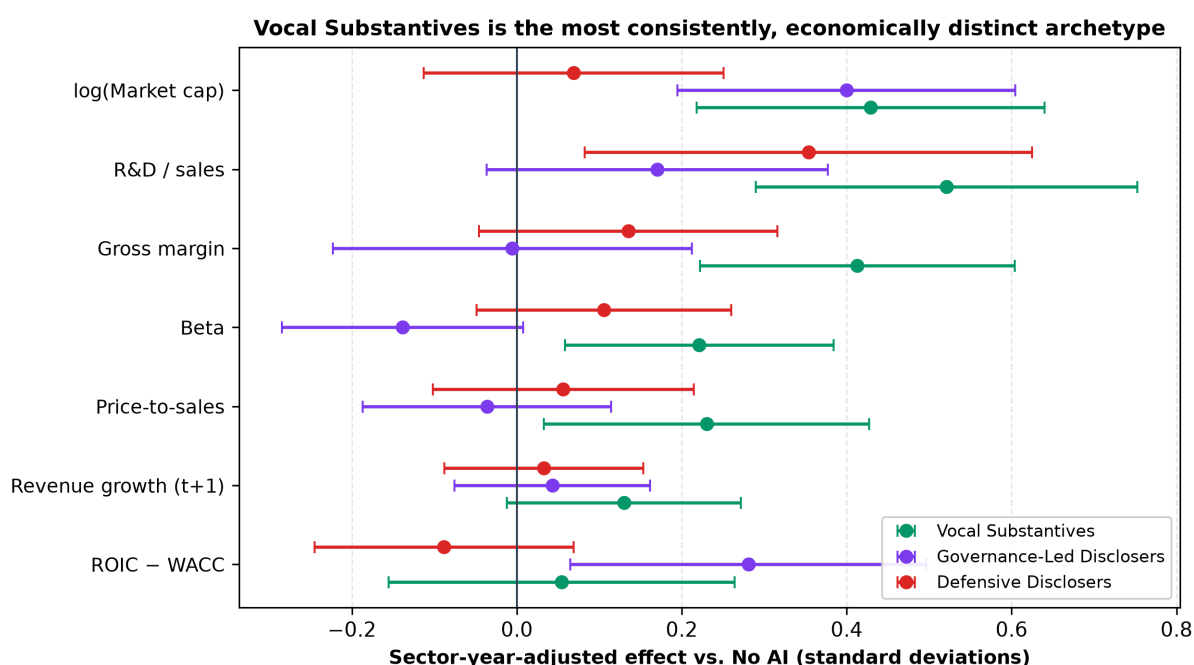


Figure 14: Sector-year-adjusted effect of disclosure archetype on firm fundamentals, relative to No AI (95% CI, firm-clustered SE). Vocal Substantives separates most consistently from the sector-year baseline.

#### 4.7.1 WHAT CHAPTERS 3 AND 4 HAVE ESTABLISHED

Together, Chapters 3 and 4 provide a comprehensive empirical map of corporate AI disclosure:

1. **Strategic Differentiation:** In an era of universal AI mention, firms separate into three stable communication postures—Vocal Substantives, Governance-Led Disclosers, and Defensive Disclosers (Chapter 3).
2. **Behavioral Realism:** These postures correspond to distinct operational activities: Vocal Substantives build customer products, Governance-Led firms automate internal workflows with governance guardrails, and Defensive firms focus on risk avoidance (Chapter 4).

3. **The Substance Deficit:** Grounding has lagged disclosure volume: only 7.7% of claims report quantified outcomes, marginal concreteness deteriorated after ChatGPT, and firms obscure vendor dependencies under proprietary branding.
4. **Economic Reality:** Vocal Substantives possess the fundamental R&D and margin profile to support aggressive technological positioning.

Different firms communicate with different postures, disclose different mixes of AI activities, and possess distinct economic profiles. Moreover, corporate AI claims vary wildly in their degree of grounding, and the GenAI boom has widened the gap between claim volume and verifiable substance. But when does this divergence between discourse and substance become a credibility problem? Chapter 5 turns directly to testing the credibility of corporate AI disclosure.

## 5. TESTING THE CREDIBILITY OF CORPORATE AI DISCLOSURE

### 5.1 WHAT CAN CREDIBILITY MEAN WHEN ACTUAL ADOPTION IS UNOBSERVED?

Without an audited census of real-world corporate AI adoption, what observable evidence makes an artificial intelligence claim more or less credible? In public company disclosures, researchers and regulators cannot verify the operational reality behind every algorithmic statement. We cannot definitively classify a disclosure as “true” or “false,” nor can this thesis identify legal fraud.

However, while credibility cannot be observed directly, it leaves observable traces. Rather than relying on mention counts or treating promotional language as an inherent indicator of deceit, this chapter evaluates corporate AI credibility through three progressively demanding empirical tests:

Table 12: Three observable credibility tests.

| Test                   | Empirical Question                                                                           | Observable Trace                                                                                           |
|------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| I. Substance           | Does narrative discourse scale with concrete, grounded activity?                             | Alignment along the 45° percentile rank diagonal of disclosure intensity vs. disclosed substance ( $W$ ).  |
| II. Consistency        | Do specific claims and evidence survive when a firm shifts institutional venues?             | Within-firm, within-period persistence of facts between informal earnings calls and statutory SEC filings. |
| III. Scrutiny response | Does corporate communication become more grounded when the cost of unsupported claims rises? | Dynamic event-study and DiD response to SEC regulatory warnings and enforcement actions (2024Q1).          |

Under this framework, “AI washing” ceases to be an arbitrary label assigned to individual firms. It is operationalized as an empirically observable hypothesis about **disclosure–substance decoupling**.

### 5.2 TEST I — DOES TALK SCALE WITH SUBSTANCE?

The first test of credibility asks whether the volume of corporate AI rhetoric is commensurate with the volume of concrete, grounded activity a firm reports. We examine the joint distribution of two separately extracted quantities across 1,449 firm-years disclosing AI (474 firms, 2021–2025):

1. **Disclosure Intensity** ( $\text{Disclosure}_{it}$ ): Semantic AI frames per 1,000 words of scorable narrative text.

2. **Disclosed Substance** ( $\text{Substance}_{it}$ ): Disclosed activity volume scaled by empirical-Bayes grounding:  $\log(1 + n_{\text{activities},it}) \times \text{Grounding}_{it}$ .

Plotting the percentile ranks of both measures against each other reveals the baseline relationship (Figure 15). Percentile ranks compress both distributions to [0,1], ensuring that technology conglomerates with hundreds of activities (e.g., Microsoft, NVIDIA) are not mechanically flagged as outliers simply due to scale.

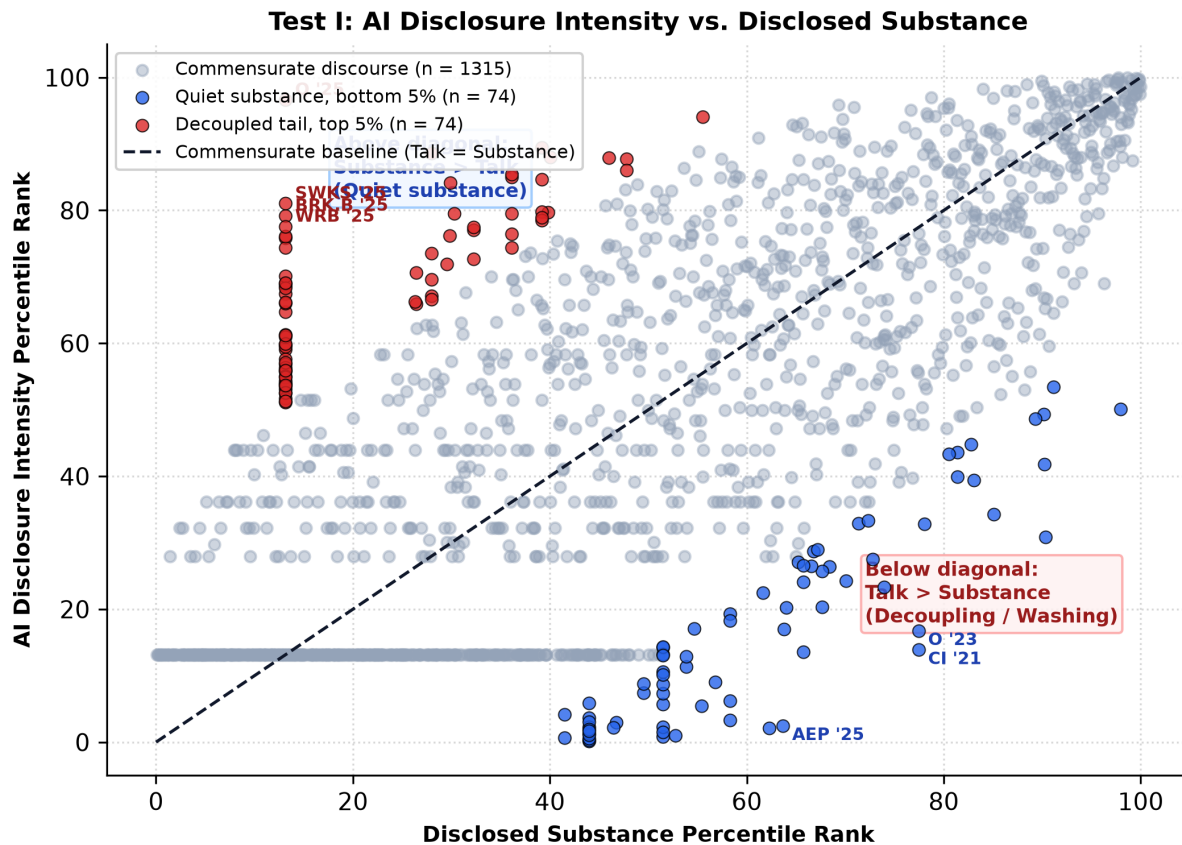


Figure 15: Test I: AI disclosure intensity vs. disclosed substance (n = 1,449 firm-years disclosing AI, 474 firms). The 45° line represents commensurate communication; firms below the diagonal exhibit disclosure running ahead of substance.

The 45° dashed line defines commensurate communication: - **Near the diagonal**: A firm's narrative intensity roughly matches its disclosed operational substance. - **Above the diagonal**: Firms that disclose substantial concrete activity while maintaining a relatively muted narrative intensity ("quiet substance"). - **Below the diagonal**: Firms whose narrative volume runs substantially ahead of their concrete disclosed activities.

The **AI-washing index** ( $W$ ) is the signed distance from this diagonal:

$$W_{it} = \text{pctrank}(\text{Disclosure}_{it}) - \text{pctrank}(\text{Substance}_{it})$$

A descriptive cut at the top 5% flags 73 firm-years in the washing tail (e.g., Q, BLK, SWKS, WRB). These are not firms with the highest AI volume in absolute terms, but firms reporting heavy promotional rhetoric with zero or near-zero grounded activities in that fiscal year.

### 5.3 IS DECOUPLING PERSISTENT OR EPISODIC?

Does a high washing score reflect a temporary, noisy year, or a persistent corporate disclosure posture? To answer this, we evaluate the year-over-year persistence of  $W_{it}$  across consecutive firm-years (Figure 16).

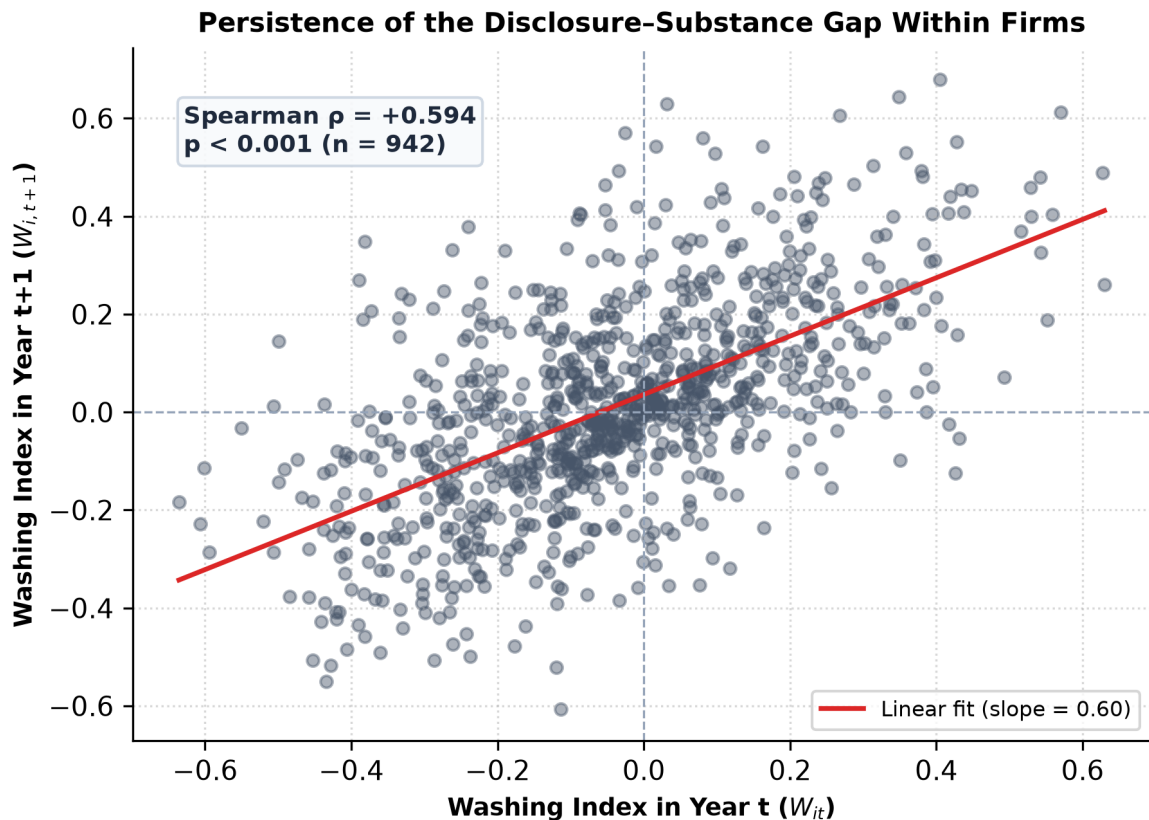


Figure 16: Year-over-year persistence of the AI-washing index ( $W_{it}$  vs.  $W_{i,t+1}$ ) across 942 consecutive firm-year pairs (Spearman rho = +0.594,  $p < 0.001$ ).

Across 942 consecutive firm-year pairs, the rank correlation is strongly positive: **Spearman**  $\rho = +0.594$  ( $p < 0.001$ ). A split-half check (averaging  $W$  across random halves of each firm's history) yields Spearman  $\rho = +0.514$  ( $p < 0.001$ ,  $n = 406$ ).

This provides a vital empirical finding: **disclosure–substance decoupling is not merely an episodic artifact of a single reporting season; it exhibits meaningful persistence within firms**. At the same time, the correlation is well below unity, confirming that  $W$  is not a static firm trait but a dynamic reporting posture that evolves over time.

### 5.4 TEST II — DOES THE CLAIM SURVIVE A CHANGE OF VENUE?

The second credibility test evaluates informational persistence across institutional venues. When the same firm communicates about the same fiscal period under two different regulatory regimes, what parts of its AI narrative survive?

We compare corporate communication across the two primary disclosure venues: 1. **Earnings Conference Calls**: Voluntary, flexible, interactive communication directed at market analysts, governed primarily by general anti-fraud provisions (Frankel, Johnson, and Skinner 1999, 133–34; Matsumoto, Pronk, and Roelofsen 2011, 1383–84). 2. **Statutory SEC Filings (10-K / 10-Q)**: Highly

structured, formal reports carrying statutory liability for material omissions and misstatements under Section 11 and Section 18 of the Exchange Act.

The empirical design incorporates **Firm × Fiscal-Year fixed effects** ( $\alpha_{i,t}$ ) with exactly two observations per cell (the call and the filing). The estimator reduces to a within-cell difference:  $\text{Call}_{i,t} - \text{Filing}_{i,t}$ . This setup automatically sweeps out all firm-specific unobservables, business models, and industry-wide macroeconomic shocks common to that fiscal period.

The extensive-margin sample comprises 2,281 firm × fiscal-year cells across 479 firms (449 present on both sides of the 2024 cutoff). Comparing the underlying activity composition across 459 cells (183 firms with  $\geq 3$  activities in each venue), a striking pattern emerges: **the activity mix itself essentially survives the venue transition**. Customer-facing deployment, internal operational deployment, and proprietary technology shares differ by only 0.6 to 2.2 percentage points between calls and filings—none statistically distinguishable from zero.

## 5.5 THE CREDIBILITY GAP IS NOT WHAT FIRMS CLAIM TO DO — IT IS WHAT EVIDENCE THEY COMMIT TO

If the mix of disclosed activities is essentially unchanged across venues, where does the credibility gap reside?

The answer lies in the **evidentiary grounding** attached to those activities ([Figure 17](#)). The earnings conference call systematically incorporates specific, falsifiable evidence that disappears when the same firm signs its statutory filing:

- **Named business function:** ↓ 11.7 percentage points in filings.
- **Quantified outcome or performance metric:** ↓ 11.0 percentage points in filings (calls state the numbers; filings omit them).
- **Named third-party provider or partner:** ↓ 5.3 percentage points in filings (nearly tripled in calls).
- **Named proprietary product or platform:** ↓ 3.8 percentage points in filings.

Conversely, statutory filings retain relatively more boilerplate governance language, regulatory caveats, and unanchored risk factors.

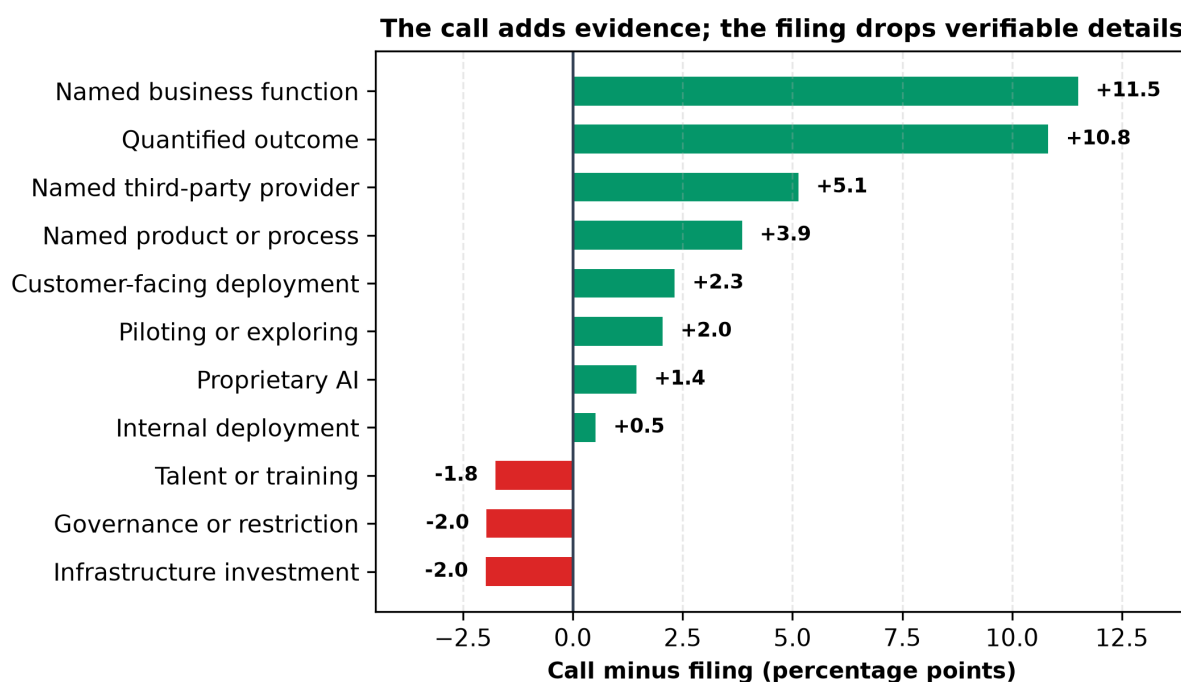


Figure 17: Same activity, different evidence: call-minus-filing gap in the proportion of disclosed activities carrying specific evidence attributes (percentage points,  $n = 459$  firm-year cells).

This constitutes the central finding of the cross-venue analysis: **the institutional venue changes less what firms say they are doing than how much evidence they attach to it**. Public companies readily pitch concrete metrics, named commercial partners, and operational gains to Wall Street analysts in earnings calls, but sanitize those same claims when drafting legally binding statutory filings.

## 5.6 PROMOTION AND CREDIBILITY ARE NOT OPPOSITES

A conventional assumption in disclosure studies is that promotional rhetoric is inversely related to credibility: that high promotion equals “cheap talk” or “washing,” while unpromotional disclosures are substantive.

Our cross-venue evidence directly refutes this simplistic dichotomy. When measuring narrative density per 1,000 words, earnings calls contain **9.9× more promotional statements** than filings—but they simultaneously contain **17.1× more quantified outcome claims** ([Table 13](#) and [Figure 18](#)).

Table 13: Channel means and call-to-filing ratio, per 1,000 words.

| Measure (per 1,000 words) | Filing | Call  | Call / Filing |
|---------------------------|--------|-------|---------------|
| Promotional claims        | 0.006  | 0.054 | 9.9×          |
| Quantified claims         | 0.003  | 0.052 | 17.1×         |

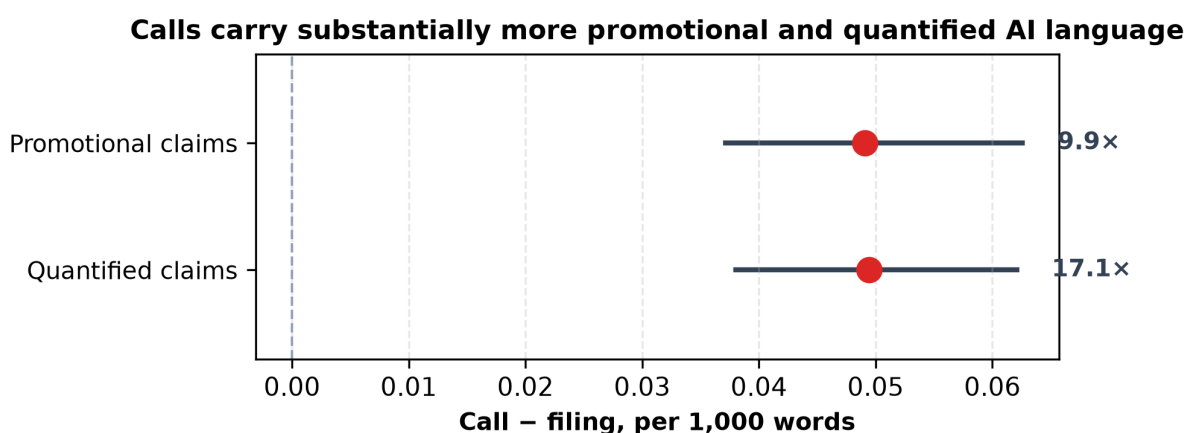


Figure 18: Within-firm, within-year call-minus-filing difference in AI disclosure rate, with a firm-clustered 95% bootstrap confidence interval. Ratios (call / filing) are annotated on the right.

Promotional communication can simultaneously be more informative. Tone alone is weak evidence of washing; what matters is whether claims carry verifiable detail and whether that detail persists across regulatory venues. This suggests a four-quadrant disclosure typology:

Table 14: Credibility matrix: promotional tone versus observable evidence.

|                | Low Observable Evidence                                                                      | High Observable Evidence                                                                    |
|----------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Low Promotion  | <i>Generic / Formal Boilerplate</i> (Standard risk factors, safe harbor lines)               | <i>Quiet Substance</i> (Operational integration disclosed without marketing hype)           |
| High Promotion | <i>Hype Candidate / Washing Risk</i> (Aspirational positioning without verifiable grounding) | <i>Promotional but Grounded</i> (Aggressive commercial pitching backed by concrete metrics) |

Earnings calls frequently operate in the “Promotional but Grounded” quadrant, while filings gravitate toward “Generic / Formal Boilerplate.”

## 5.7 TEST III — DOES SCRUTINY CHANGE CREDIBILITY?

The third and most demanding test of credibility asks: **if regulatory scrutiny increases the potential cost of ungrounded claims, do firms adjust their disclosure posture to become more grounded or more consistent?**

In February 2024, SEC Chair Gensler issued an explicit public warning on “AI washing,” followed by enforcement actions against investment advisers in March 2024 and comment letters to public registrants (2024Q1 cutoff). We test whether firms with higher pre-event decoupling ( $W$ ) responded to this scrutiny.

In the continuous event-study specification (5,376 firm-quarters across 490 firms), pre-trends for promotional intensity ( $F = 1.73, p = 0.142$ ) and specificity ( $F = 1.25, p = 0.289$ ) remain flat prior to 2024Q1. However, post-event coefficients reveal no significant differential change in promotional language (+0.0025,  $p = 0.416$ ) or specificity (+0.0022,  $p = 0.905$ ), while risk disclosure rises (+0.0070,  $p = 0.021$ ).

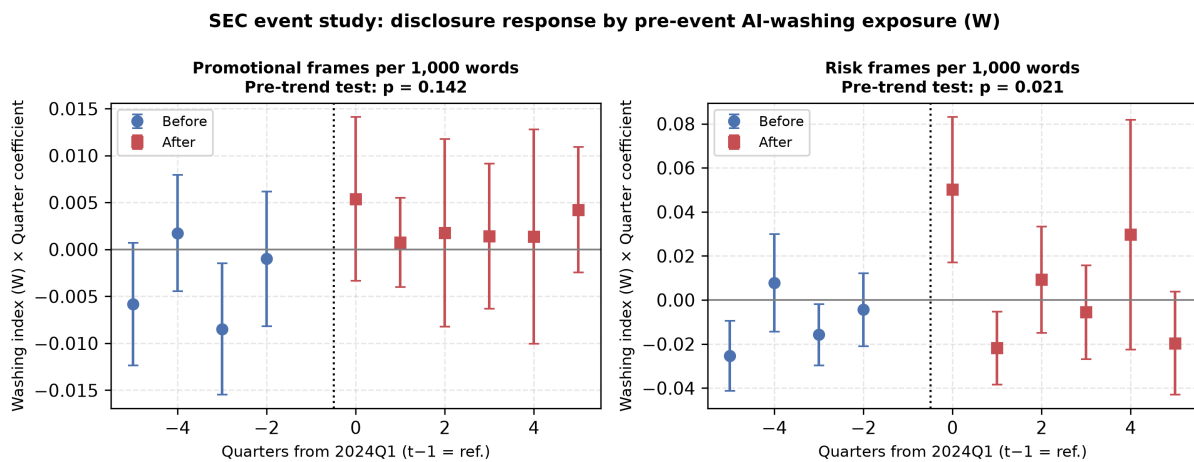


Figure 19: Diagnostic event study: SEC regulatory sweep around AI disclosures across event quarters, using pre-event AI-washing index (W) as treatment exposure.

A difference-in-differences design splitting firms by pre-2024 washing score (132 high-washing treated firms vs. 358 controls) shows that high-washing firms differentially increased specific AI disclosures (+0.0103 per 1,000 words,  $p < 0.001$ ) and quantified metrics (+0.0060,  $p = 0.017$ ; pre-trend  $p = 0.250$ , [Figure 20](#)).

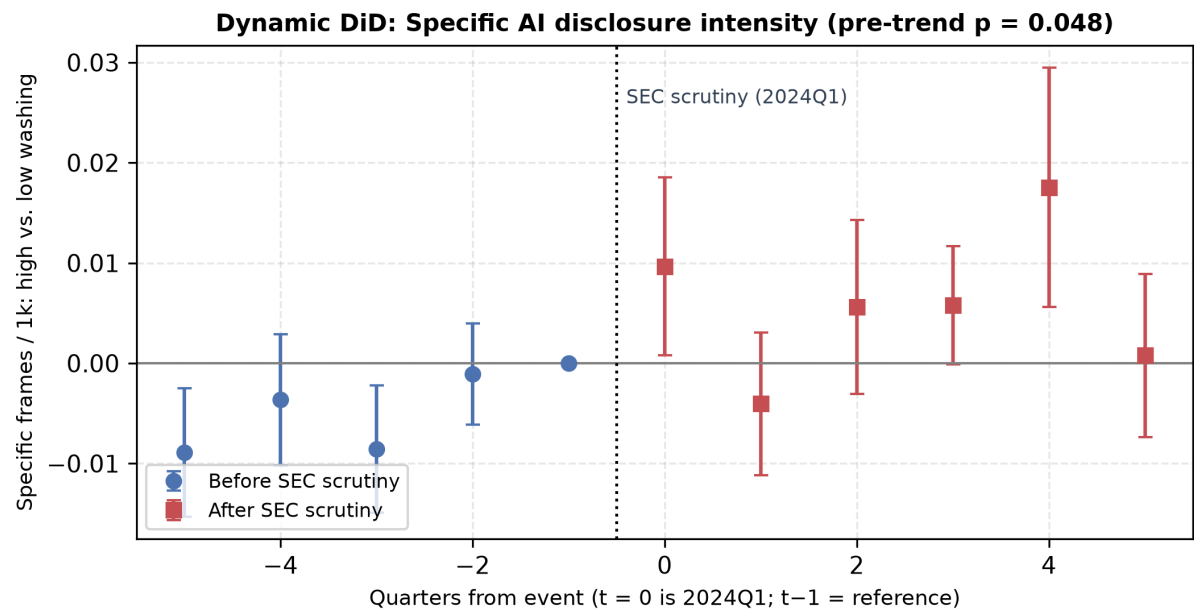


Figure 20: Difference-in-differences dynamic trajectory: specific disclosure intensity per 1,000 words following SEC regulatory scrutiny, comparing high vs. low pre-event AI-washing firms.

Furthermore, testing the cross-venue gap itself reveals that the promotional gap widened from 0.019 in 2021–2022 to 0.077 in 2024 and 0.080 in 2025. Pre-trend tests fail ( $p < 0.01$ ) because the gap was already expanding in fiscal 2023, well before SEC enforcement.

**Identification verdict: there is no credible evidence that regulatory scrutiny closed the credibility gap.** While high-washing firms added more specificity and governance after December 2023, the shift

concentrates heavily in the 2024 annual filing season rather than tracking enforcement dates. What is observed in the filings is equally consistent with industry-wide generative AI maturation.

Table 15: SEC identification verdict.

| Design                                                                    | Pre-trend test                                             | Verdict                                           |
|---------------------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------|
| Event study, continuous washing index exposure ( $W$ ), firm + quarter FE | Passes for promo ( $p = 0.142$ ) and spec ( $p = 0.289$ )  | Null differential post shift ( $p > 0.40$ )       |
| Simple DiD, pre-event washing index split ( $W > \text{median}$ )         | Passes for promo ( $p = 0.230$ ) and quant ( $p = 0.250$ ) | Specificity jump concentrated in 2024 10-K season |
| Within-firm channel gap (call – filing), post-2024                        | Fails ( $p < 0.01$ ); gap was already widening in FY2023   | Gap persists; no closure after scrutiny           |
| DeepSeek event study (Appendix G)                                         | Passes for promo ( $p = 0.715$ ) and spec ( $p = 0.638$ )  | Null post effect ( $p > 0.20$ )                   |

## 5.8 WHAT CAN — AND CANNOT — BE CALLED AI WASHING?

Synthesizing the three credibility tests establishes the empirical boundary of what can—and cannot—be called AI washing:

### 5.8.1 WHAT OUR DATA EMPIRICALLY ESTABLISH:

1. **Decoupling exists:** In a meaningful subset of firms, disclosure intensity runs substantially ahead of concrete, disclosed operational substance ( $W > 0$ ).
2. **Decoupling is persistent:** A firm's position in the washing index is stable over time ( $\rho \approx 0.60$ ), reflecting an enduring communication posture rather than episodic noise.
3. **Evidence does not survive across venues:** Public firms attach quantifiable metrics, business functions, and partner attributions to earnings calls, but omit them from monitored statutory filings.
4. **Scrutiny did not eliminate the gap:** Regulatory warnings prompted risk factor expansion and selective specificity in annual reports, but the 9.9× promotional cross-venue gap remains wide open.

### 5.8.2 WHAT OUR DATA CANNOT ESTABLISH:

- We cannot claim that any individual corporate statement is legally fraudulent or intentionally misleading.
- We cannot prove that an ungrounded activity is non-existent in reality, because firms may deploy proprietary algorithms without disclosing technical details.
- We cannot interpret third-party vendor opacity as deliberate concealment.
- We cannot interpret  $W$  as a measure of fraud.

Therefore:

**In this thesis, “AI washing” is not a verdict of false implementation. It is an empirically observable form of disclosure—substance decoupling.**

## 6. ECONOMIC AND MARKET RELEVANCE OF AI DISCLOSURE

This chapter investigates whether capital markets reprice the *volume* of corporate AI rhetoric or the *substance* of operational execution around earnings calls. Tracking communication call-by-call across S&P 500 firms from 2021 to 2025, we test how post-call market beta responds to established disclosure postures, call-specific narrative surprises, and concrete operational activities.

### 6.1 EMPIRICAL STRATEGY AND RESEARCH DESIGN

The unit of observation is a firm-date earnings call event. Multiple transcripts occurring on the same day are aggregated, and calls with zero mentions are retained as valid empirical zeros. The call date anchors the timeline:

- **Historical Posture** ( $\text{HistD}_{i,t-}, \text{HistS}_{i,t-}$ ): Expanding mean of disclosure and substance across calls strictly prior to date  $t$ .
- **Call-Specific Surprise** ( $\text{SurpriseD}_{i,t}, \text{SurpriseS}_{i,t}$ ): Current call value minus the firm's historical mean ( $\text{Value}_{i,t} - \text{Hist}_{i,t-}$ ).
- **Disclosure Intensity**: AI frames per 1,000 words.
- **Substance**:  $\log(1 + \text{Activities}/1,000 \text{ words}) \times \text{Grounding}$ , where grounding is the proportion of activities tied to concrete business operations.

Pre- and post-call market betas are estimated over non-overlapping windows to avoid look-ahead bias and earnings-announcement drift: -  $\beta_{i,t}^{\text{pre}}$  over  $[-252, -21]$  trading days before the call. -  $\beta_{i,t}^{\text{post}}$  over  $[+21, +126]$  trading days after the call.

Predetermined controls ( $X_{i,t}^{\text{pre}}$ ) include log market cap, 60-day return momentum ( $\text{Return}_{60}$ ), and pre-call idiosyncratic volatility. Point-in-time accounting fundamentals (ROA, leverage) are linked from the most recent 10-K or 10-Q filed strictly before the call ( $\text{filing\_date} < \text{call\_date}$ ), extracted from XBRL accession facts.

### 6.2 INCREMENTAL INFORMATION OF AI DECOUPLING

Before evaluating call-by-call return and beta dynamics, we test whether the firm-year AI-washing index ( $W$ ) adds explanatory power beyond disclosure volume and substantive activity. Adding  $W$  to the baseline beta specification raises  $R^2$  from 0.1732 to 0.1735 ( $n = 711$ ;  $\Delta R^2 = +0.0004$ ; Wald  $F = 0.12, p = 0.725$ ).

While the credibility gap is persistent and managerially interpretable within Chapter 5's disclosure framework, it carries essentially zero incremental information about systematic equity risk once underlying specificity and concrete activity are already in the model.

### 6.3 BASELINE ECONOMETRIC MODEL

We estimate an ANCOVA specification conditioning on pre-call beta:

$$\beta_{i,t}^{\text{post}} = \alpha + \beta_1 \text{HistD}_{i,t-} + \beta_2 \text{HistS}_{i,t-} + \beta_3 \text{SurpriseD}_{i,t} + \beta_4 \text{SurpriseS}_{i,t} + \gamma \beta_{i,t}^{\text{pre}} + \delta X_{i,t}^{\text{pre}} + \alpha_{s,y} + \epsilon_{i,t},$$

with industry-by-year fixed effects ( $\alpha_{s,y} \equiv \text{SIC2} \times \text{Year}$ ) and standard errors clustered by firm. Continuous regressors are standardized to zero mean and unit variance.

[Table 16](#) and [Figure 21](#) report the baseline estimates.

Table 16: Baseline earnings-call beta regression: standardized coefficients, 95% confidence intervals clustered by firm, and p-values.

| Variable                          | Beta est. | IC95%             | p     |
|-----------------------------------|-----------|-------------------|-------|
| Historical disclosure intensity   | 0.0701    | [0.0207, 0.1196]  | 0.005 |
| Historical substantive activity   | 0.0082    | [-0.0440, 0.0603] | 0.759 |
| Call-specific disclosure surprise | 0.0286    | [0.0044, 0.0528]  | 0.02  |
| Call-specific substance surprise  | 0.0029    | [-0.0175, 0.0232] | 0.783 |

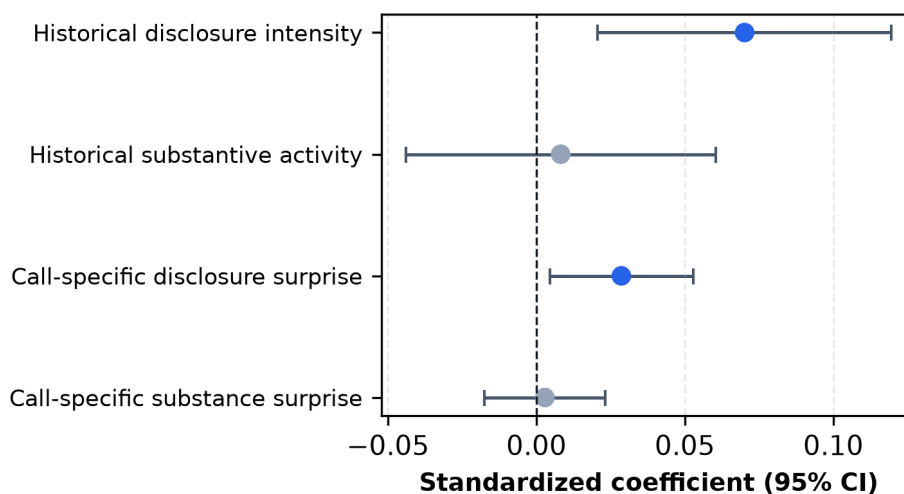


Figure 21: Standardized coefficients and 95% confidence intervals for disclosure and substantive activity in the baseline post-call beta model.

The baseline estimation covers 6,794 calls across 447 firms (262 industry-by-year cells,  $R^2 = 0.692$ ). The median firm has 9.0 prior calls in the panel, ensuring well-anchored historical averages.

Two main patterns emerge:

1. **Voice matters:** Historical disclosure intensity is positive and significant ( $\beta = 0.070, p = 0.005$ ; unstandardized 0.103 per 1 SD increase), as is call-specific surprise disclosure ( $\beta = 0.029, p = 0.020$ ; unstandardized 0.030). Both established rhetorical posture and unexpected narrative spikes raise subsequent systematic risk.
2. **Substance does not:** Historical substance ( $\beta = 0.008, p = 0.759$ ) and surprise substance ( $\beta = 0.003, p = 0.783$ ) are indistinguishable from zero. Joint tests confirm this contrast: the disclosure block (HistD + SurpriseD) is strongly significant ( $F = 5.15, p = 0.006; \Delta R^2 = +0.0051$ ), whereas the substance block is null ( $F = 0.05, p = 0.952$ ).

The market prices executive disclosure volume, but remains indifferent to grounded operational activities.

## 6.4 ROBUSTNESS AND SENSITIVITY ANALYSIS

[Table 17](#) compares the baseline ANCOVA model against specifications incorporating accounting controls, firm fixed effects, and first-differencing.

Table 17: Robustness specifications for earnings-call beta regressions.

| Variable                 | Baseline         | + Accounting      | Firm FE / within  | Delta beta       |
|--------------------------|------------------|-------------------|-------------------|------------------|
| HistDisclosure           | 0.0707 (p=0.004) | 0.0633 (p=0.021)  | 0.0273 (p=0.589)  | 0.0203 (p=0.321) |
| HistSubstance            | 0.0071 (p=0.788) | 0.0168 (p=0.595)  | -0.0947 (p=0.035) | 0.0192 (p=0.409) |
| SurpriseDisclosure       | 0.0275 (p=0.026) | 0.0275 (p=0.065)  | 0.0101 (p=0.442)  | 0.0161 (p=0.382) |
| SurpriseSubstance        | 0.0027 (p=0.794) | -0.0004 (p=0.976) | -0.0181 (p=0.106) | 0.0174 (p=0.265) |
| Beta pre                 | ✓                | ✓                 | ✓                 | —                |
| Market controls          | ✓                | ✓                 | ✓                 | ✓                |
| ROA + liabilities/assets | —                | ✓                 | —                 | —                |
| SIC2 × CallYear FE       | ✓                | ✓                 | —                 | ✓                |
| Firm FE                  | —                | —                 | ✓                 | —                |
| N                        | 6,794            | 4,580             | 6,794             | 6,794            |

#### 6.4.1 ESTIMATION WINDOW DYNAMICS

To evaluate whether this repricing reflects transient post-announcement volatility or a persistent risk adjustment, we re-estimate the model across alternative post-call windows ([+21, +63], [+21, +126], [+21, +252]) and pre-call windows ([-126, -21], [-252, -42]).

[Table 18](#) reports standardized coefficients across all five horizons.

Table 18: Sensitivity of post-call beta regressions across alternative post-call and pre-call estimation horizons. Standardized coefficients are reported with p-values clustered at the firm level.

| Variable           | [+21, +63]        | [+21, +126] (Base) | [+21, +252]       | Pre [-126, -21]  | Pre [-252, -42]  |
|--------------------|-------------------|--------------------|-------------------|------------------|------------------|
| HistDisclosure     | 0.0822 (p<0.001)  | 0.0707 (p=0.004)   | 0.0762 (p=0.005)  | 0.0939 (p=0.001) | 0.0733 (p=0.004) |
| HistSubstance      | -0.0166 (p=0.493) | 0.0071 (p=0.788)   | 0.0037 (p=0.901)  | 0.0034 (p=0.915) | 0.0069 (p=0.804) |
| SurpriseDisclosure | 0.0270 (p=0.089)  | 0.0275 (p=0.026)   | 0.0355 (p=0.005)  | 0.0362 (p=0.007) | 0.0295 (p=0.020) |
| SurpriseSubstance  | -0.0002 (p=0.986) | 0.0027 (p=0.794)   | -0.0024 (p=0.820) | 0.0006 (p=0.953) | 0.0022 (p=0.839) |

Three dynamics stand out:

- **Historical posture persists:** HistDisclosure remains positive and significant across all five windows ( $\beta \in [0.071, 0.094]$ ,  $p \leq 0.004$ ), sustaining its pricing impact through a full trading year.
- **Surprise disclosure compounds:** SurpriseDisclosure strengthens monotonically from +63 days ( $\beta = 0.027$ ,  $p = 0.089$ ) to +126 days ( $\beta = 0.028$ ,  $p = 0.026$ ) and +252 days ( $\beta = 0.036$ ,  $p = 0.005$ ). Narrative shocks are not transient noise; they are gradually integrated into forward risk expectations.
- **Substance is uniformly null:** Both historical and surprise substance stay near zero across all horizons ( $p > 0.49$ ).

#### 6.4.2 MODEL SPECIFICATIONS AND IDENTIFICATION

- **Accounting fundamentals:** In the matched sample with observed XBRL fundamentals ( $N = 4,580$ ), adding ROA and liabilities-to-assets leaves disclosure estimates intact (HistD = 0.063,  $p = 0.021$ ; Surprised = 0.028,  $p = 0.065$ ). The signal is not an artifact of omitted profitability or leverage.
- **First-differences (DeltaBeta):** When modeling raw beta changes ( $\beta^{\text{post}} - \beta^{\text{pre}}$ ), all AI terms become insignificant ( $p \in [0.27, 0.41]$ ). Because market beta mean-reverts ( $\gamma \approx 0.58 < 1.0$ ),

forcing  $\gamma = 1$  introduces mechanical attenuation bias, confirming ANCOVA as the appropriate specification.

- **Between-firm sorting vs. within-firm identification:** Under firm fixed effects, HistDisclosure attenuates ( $\beta = 0.027, p = 0.589$ ), while HistSubstance turns negative ( $\beta = -0.095, p = 0.035$ ). The baseline is driven by cross-sectional sorting—vocal AI adopters carry higher systematic risk than peers—rather than high-frequency quarterly shifts within the same firm.

### 6.4.3 DIAGNOSTICS AND SUBSETS

- **Collinearity:** VIFs remain below 3.8 (well beneath the threshold of 5.0), confirming standard errors are not inflated by collinearity despite the correlation between disclosure and substance ( $\rho = 0.84$ ).
- **Standalone subsets:** Historical disclosure alone yields  $\beta = 0.083$  ( $p = 0.003$ ); surprise disclosure alone yields  $\beta = 0.055$  ( $p < 0.001$ ).
- **Influential observations:** Winsorizing at 1/99% preserves both disclosure terms ( $p \leq 0.048$ ). Trimming 1% preserves HistDisclosure ( $\beta = 0.084, p < 0.001$ ) while attenuating SurpriseDisclosure ( $\beta = 0.008, p = 0.525$ ), showing that surprises partly reflect large narrative shocks.
- **Factor benchmark:** Fama-French 3-factor betas yield identical conclusions (HistD = 0.062,  $p = 0.014$ ; SurpriseD = 0.047,  $p = 0.002$ ; substance null).

## 6.5 LEVERAGE ROBUSTNESS: MATCHED SAMPLES

To test whether capital structure confounds the disclosure signal, Models 0 and 1 are estimated on the exact subsamples with observed point-in-time XBRL leverage ratios:

$$\text{Model 0: } \beta_{i,t}^{\text{post}} = \alpha + \beta_1 \text{HistD}_{i,t-} + \beta_2 \text{HistS}_{i,t-} + \beta_3 \text{SurpriseD}_{i,t} + \beta_4 \text{SurpriseS}_{i,t} + \gamma \beta_{i,t}^{\text{pre}} + \delta X_{i,t}^{\text{pre}} + \alpha_{s,y} + \epsilon_{i,t}$$

$$\text{Model 1: } \beta_{i,t}^{\text{post}} = \text{Model 0} + \theta \text{Leverage}_{i,t} + \epsilon_{i,t}$$

### 6.5.1 LONG-TERM DEBT / EQUITY

[Table 19](#) reports results for Debt / Equity ( $N = 3,223$ ).

Table 19: Matched-sample leverage robustness: Debt / Equity.

| Variable                      | M0     | M1     |
|-------------------------------|--------|--------|
| HistDisclosure, beta est.     | 0.0532 | 0.0533 |
| HistDisclosure, p             | 0.111  | 0.111  |
| SurpriseDisclosure, beta est. | 0.0344 | 0.0344 |
| SurpriseDisclosure, p         | 0.074  | 0.074  |
| Leverage, beta est.           | —      | 0.0027 |
| Leverage, p                   | —      | 0.599  |
| R <sup>2</sup>                | 0.7290 | 0.729  |

Adding Debt / Equity leaves disclosure coefficients identical (HistD = 0.053, SurpriseD = 0.034), with leverage explaining zero incremental variance ( $p = 0.599$ ).

### 6.5.2 TOTAL LIABILITIES / TOTAL ASSETS

[Table 20](#) reports results using Total Liabilities / Total Assets ( $N = 3,255$ ).

Table 20: Matched-sample leverage robustness: Liabilities / Assets.

| Variable                      | M0     | M1      |
|-------------------------------|--------|---------|
| HistDisclosure, beta est.     | 0.0624 | 0.0577  |
| HistDisclosure, p             | 0.045  | 0.06    |
| SurpriseDisclosure, beta est. | 0.0313 | 0.0298  |
| SurpriseDisclosure, p         | 0.072  | 0.082   |
| Leverage, beta est.           | —      | -0.0383 |
| Leverage, p                   | —      | 0.008   |
| R <sup>2</sup>                | 0.7103 | 0.7113  |

Controlling for liabilities-to-assets results in only minor coefficient shifts (HistD moves from 0.062 to 0.058; Surprised from 0.031 to 0.030). While significant ( $\beta = -0.038, p = 0.008$ ), leverage adds under 0.1 percentage point to  $R^2$  (0.7103 to 0.7113) without displacing the disclosure terms.

## 6.6 CRITICAL LIMITATIONS

1. **Temporal concentration in the 2024 AI bubble:** Leave-one-year-out regressions remain significant when dropping 2021, 2022, 2023, or 2025 ( $p < 0.03$ ). However, excluding 2024 collapses the result: HistDisclosure drops to 0.026 ( $p = 0.136$ ) and SurpriseDisclosure to 0.004 ( $p = 0.732$ ). The repricing of call disclosures is historically anchored in the 2024 generative AI wave.
2. **Pre-call beta placebo failure:** Regressing pre-call beta ( $\beta_{[-252, -21]}^{\text{pre}}$ ) on contemporaneous call disclosure yields  $\beta = 0.176$  ( $p < 0.001$ ), while substance is null ( $p = 0.196$ ). Firms that talk heavily about AI were already high-beta firms prior to the call. HistDisclosure therefore reflects a persistent corporate posture adopted by high-beta, growth-oriented firms seeking market attention, rather than an exogenous shock to systematic risk.

## 7. DISCUSSION AND MANAGERIAL IMPLICATIONS

The results support three complementary interpretations. Specific, falsifiable claims fit a signalling account; the disappearance of the no-AI segment and the limited economic distinctiveness of most archetypes fit institutional diffusion; and the widening call-filing gap is consistent with cheap talk in a more discretionary, lower-formality disclosure venue. These are associations, not evidence that one mechanism causally governs all AI disclosure.

### 7.1 ISSUER IMPLICATIONS

The measurement built across this thesis converts directly into a disclosure checklist. Credible AI disclosure:

- names a concrete activity, not “AI capabilities,” which 86% of firms use as an empty placeholder object;
- states a clear stage (deployed or scaled, not merely explored or piloted);
- names a specific product or process;
- reports a quantified outcome;
- discloses the source of the AI, including provider and partner relationships, since 16% of activities in the corpus leave this unstated, and undeclared origin is far more common among Defensive Disclosers and Governance-Led Disclosers (33% each) than among Vocal Substantives (10%); and

- is consistent between what is said on the earnings call and what is filed with the SEC. The Welltower comment letter illustrates why that comparison can be useful: SEC staff referenced statements across the filing, call, and press release, without finding a violation or defining a general consistency test.

## 7.2 INVESTOR IMPLICATIONS

In earnings calls, disclosure intensity carries significant information about post-call market beta, while substantive activity does not (Chapter 6).

Investors should read earnings-call AI claims against what the same firm has filed: the same activity mix is described in both venues, and the call does supply concrete evidence the filing typically omits (quantified outcomes, named providers, named products) alongside more promotional language, so cross-venue consistency, not the call's promotional tone by itself, is the more useful credibility signal. This is a gap that has been widening since the 2023 AI boom and has shown no sign of narrowing since the SEC's December 2023 warning.

The within-disclosure index is best read through decoupling theory: formal disclosure can run ahead of the concrete activity a firm discloses, without establishing what the firm actually does. The cross-venue gap is the closer analogue to the mechanism the SEC has enforced. Neither measure has an external operational benchmark, and the SEC design does not identify a causal effect of scrutiny.

Table 21: Issuer / investor checklist.

| Dimension             | Issuer guidance                                          | Investor read                                                                      |
|-----------------------|----------------------------------------------------------|------------------------------------------------------------------------------------|
| Named activity        | State the specific action-object, not "AI capabilities"  | Absence is the single most common gap (86% of firms)                               |
| Stage                 | Report deployed/scaled, not only piloted                 | Piloting-only claims should be discounted relative to deployed claims              |
| Product/process named | Name it                                                  | Named claims carry more of the incremental economic signal                         |
| Quantified outcome    | Report the figure in the filing, not only on the call    | The filing omits this far more often than the call (61%→37% pooled-to-filing-only) |
| AI source / provider  | Disclose whether AI is own, third-party, or co-developed | 16% leave this unstated; undeclared origin clusters in the least-grounded segment  |
| Venue consistency     | Match filing content to call content                     | The 9.9× promotional gap has not closed since 2024 scrutiny                        |

## 8. CONCLUSIONS

This thesis follows a single progression: AI disclosure became nearly universal; its content differs systematically across firms; those differences reveal meaningful gaps between talk and disclosed substance, especially across disclosure venues; and finally, we ask whether these distinctions carry economic information—finding that some do, although the evidence is associative rather than causal.

First, AI disclosure grew nearly seven-fold in intensity and reached 92% of S&P 500 firms between 2021 and 2025. Firms do not merely differ in volume: their statements organize into reproducible disclosure profiles, but only Vocal Substantives—those reporting the greatest concrete activity even after conditioning on volume—show a broad, sector-year-adjusted economic distinction.

Second, credibility is institutionally contingent. The within-disclosure washing index captures a stable firm-year gap between disclosure intensity and grounded substance, while cross-venue comparisons reveal that the same firm-year places 9.9× more promotional claims and 17.1× more quantified

claims in earnings calls than in statutory filings. This venue gap widened during the generative AI surge and showed no detectable closure after 2024 SEC scrutiny.

Third, semantic content and disclosed activity relate to economic and market outcomes, but the evidence is associative rather than causal. In earnings-call beta regressions, historical disclosure posture predicts higher post-call systematic risk, but this reflects a persistent trait of high-beta firms rather than an exogenous communication shock. Once underlying specificity and activity are observed, the washing index adds no incremental explanatory power.

The methodological contribution is a five-layer, largely reproducible extraction pipeline that recovers structured, falsifiable claims and grounded activities from over nine million paragraphs at a cost no purely human-coded design could match, with venue-specific measurement error documented rather than assumed away. The managerial contribution is a concrete disclosure checklist for issuers and a cross-venue credibility framework for investors, both grounded in what the measurement shows carries information.

Theoretically, the results are most consistent with a signalling account of specificity, a legitimacy/institutional-diffusion account of the “no AI” segment’s disappearance, and a cheap-talk account of earnings calls as a more discretionary, lower-formality disclosure venue. The main claims of this thesis are measurement and association rather than causal effects: the most tightly identified design isolates a venue-associated disclosure difference, not the causal effect of legal liability, and the SEC and DeepSeek shock designs fail pre-trend tests and are reported as failed identification rather than assigned a causal reading.

Four limitations are carried through every chapter:

1. Every frame and activity label comes from an LLM judge: the measured reliability ( $\kappa$  0.81–0.87) is agreement between two LLM judges, not accuracy against an independent ground truth.
2. “Disclosed behaviour” and “disclosed activity” measure what a firm says it does, not what an audited operational record would confirm.
3. The SEC and DeepSeek shocks are not identifiable against the background AI boom with the designs available, a null result reported rather than concealed.
4. The AI-washing index has no external benchmark: it compares a firm’s own disclosure to its own disclosed activity, both extracted from the same corpus, and cannot check either against an audited operational record.

Cross-venue results are robust to using words rather than paragraphs as the denominator and to denominator-free activity comparisons.

Future research should extend the cross-venue design to a synthetic-control or matched-donor-pool setting if a denser call-transcript series becomes available, particularly for cases like Welltower with too little pre-period call content to estimate a firm-specific trajectory; and extend the comparative international panels (Chile, Italy) noted in the data section to test whether the venue-gap and segmentation results generalize outside a single, SEC-regulated regime.

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# APPENDICES

## APPENDIX A: ANALYTICAL STACK

The pipeline and analysis were implemented in Python. Parquet files are the primary storage format and DuckDB provides queryable views over those files. pandas and NumPy handle panel construction and numerical transformations; scikit-learn, SciPy, and statsmodels support the prefilter, clustering, diagnostics, and cluster-robust regressions; and matplotlib produces the figures.

Text embeddings use the Hugging Face model BAAI/bge-m3 (1,024-dimensional dense vectors) for semantic-anchor features and retrieval. The principal LLM judge is qwen/qwen3.7-flash, called through OpenRouter with structured outputs via pydantic-ai, for semantic-frame and disclosed-activity extraction. Gemini is used only as an independent judge in the agreement checks, not as the production extractor.

GPU embedding and document-processing workloads were run on a Vast.ai instance with an NVIDIA RTX 5090, using a custom-built mrbungie/datascience-runtime image. Other analytical tasks ran locally on a MacBook Pro with an M2 Pro chip. Code was synchronized with Git and data artifacts with Backblaze. SEC filings and XBRL facts are collected through EDGAR tooling. Part of the financial and market data, including historical prices, was obtained through yfinance; Fama–French factors supply the market and risk-free-rate series used in the risk measures.

## APPENDIX B: DATA PIPELINE

The data pipeline has five stages: collect and segment filings and calls; deduplicate scorable text; identify AI-relevant text with the prefilter; extract semantic frames; and extract firm-specific disclosed activities. Each stage writes a parquet artifact consumed by the next, while the DuckDB file exposes views over the current parquet data rather than a second copy. The current corpus totals in Chapter 2 and the form-by-form table are generated from the processed paragraphs table at render time.

## APPENDIX C: SUPPLEMENTARY DATA

### 8.0.1 AI DISCLOSURE STRATEGY DIMENSIONS: CORRELATION AND PCA DIAGNOSTICS

The correlation structure across the seven AI disclosure strategy dimensions is reported in [Table 22](#) below:

Table 22: Correlation matrix of the seven posture dimensions.

|             | Promotional | Risk  | Governance | Temporal (realized) | Customer-facing | Specificity | Intensity |
|-------------|-------------|-------|------------|---------------------|-----------------|-------------|-----------|
| Promotional | 1           | -0.4  | 0          | 0.17                | 0.29            | 0.28        | 0.32      |
| Risk        | -0.4        | 1     | -0.11      | -0.47               | -0.48           | -0.59       | -0.36     |
| Governance  | 0           | -0.11 | 1          | 0.23                | -0.36           | -0.07       | -0.2      |

|                     | Promotional | Risk  | Governance | Temporal (realized) | Customer-facing | Specificity | Intensity |
|---------------------|-------------|-------|------------|---------------------|-----------------|-------------|-----------|
| Temporal (realized) | 0.17        | -0.47 | 0.23       | 1                   | 0.27            | 0.44        | 0.23      |
| Customer-facing     | 0.29        | -0.48 | -0.36      | 0.27                | 1               | 0.61        | 0.59      |
| Specificity         | 0.28        | -0.59 | -0.07      | 0.44                | 0.61            | 1           | 0.39      |
| Intensity           | 0.32        | -0.36 | -0.2       | 0.23                | 0.59            | 0.39        | 1         |

A principal-components analysis of these seven standardized dimensions is reported in [Table 23](#) as a robustness diagnostic:

Table 23: PCA diagnostic on the seven posture dimensions: variance explained, reported as a robustness check, not as the clustering method.

| Component | Variance explained | Cumulative |
|-----------|--------------------|------------|
| PC1       | 43.2%              | 43.2%      |
| PC2       | 20.0%              | 63.2%      |
| PC3       | 12.1%              | 75.3%      |
| PC4       | 8.4%               | 83.7%      |
| PC5       | 7.0%               | 90.7%      |
| PC6       | 5.3%               | 96.0%      |
| PC7       | 4.0%               | 100.0%     |

## 8.0.2 FULL DISCLOSED ACTIVITIES RANKING AND ARCHETYPE PROFILES

The full ranking of action × object disclosed corporate activities in the S&P 500 ([Table 24](#) below) complements the top-10 chart already shown in Chapter 3 (“What firms disclose doing”).

Table 24: Full action x object ranking, pooled vs. filings-only share of firms, with literal textual examples.

| Action x Object family                             | % Firms (Pooled) | % Firms (Filings only) | Literal textual examples                                                       |
|----------------------------------------------------|------------------|------------------------|--------------------------------------------------------------------------------|
| deploy · AI, unspecified object                    | 88.2%            | 82.1%                  |                                                                                |
| deploy · product feature or service                | 60.5%            | 42.0%                  | ai engine, digital solutions, interconnect products, advertising tools         |
| deploy · other                                     | 55.3%            | 33.4%                  | on-device ai, edge ai, proprietary ai, aip                                     |
| deploy · ML or predictive model                    | 49.2%            | 27.1%                  | ai models, machine learning models, ai algorithms, machine learning algorithms |
| deploy · analytics or data platform                | 48.7%            | 30.7%                  | ai platform, ai platforms, ai-driven expert platform, expert platform          |
| pilot_or_explore · AI, unspecified object          | 42.0%            | 26.5%                  |                                                                                |
| scale · AI, unspecified object                     | 38.9%            | 27.3%                  |                                                                                |
| deploy · named product or platform                 | 36.8%            | 23.1%                  | aiops, aip, embedded ai, mist ai                                               |
| deploy · automation                                | 36.3%            | 19.1%                  | automation, ai and automation, automation tools, automation and ai             |
| invest_infrastructure · AI, unspecified object     | 34.9%            | 25.8%                  |                                                                                |
| hire_or_train · talent                             | 30.7%            | 26.7%                  | ai talent, talent, employees, technical talent                                 |
| integrate · AI, unspecified object                 | 28.8%            | 21.8%                  |                                                                                |
| invest_infrastructure · compute and infrastructure | 27.9%            | 17.9%                  | ai infrastructure, data centers, compute capacity, cloud infrastructure        |

| Action x Object family                     | % Firms (Pooled) | % Firms (Filings only) | Literal textual examples                                                                         |
|--------------------------------------------|------------------|------------------------|--------------------------------------------------------------------------------------------------|
| scale · other                              | 25.4%            | 9.9%                   | ai integration, hybrid cloud and ai strategy, edge ai, ai portfolio                              |
| scale · product feature or service         | 25.0%            | 9.5%                   | digital capabilities, data ecosystem, efx.ai capabilities, firefly services                      |
| measure_outcome · other                    | 23.1%            | 10.7%                  | ai orders, ai demand, business performance, inference performance                                |
| develop · AI, unspecified object           | 22.3%            | 18.7%                  |                                                                                                  |
| govern_or_control · AI, unspecified object | 21.2%            | 19.3%                  |                                                                                                  |
| invest_infrastructure · other              | 21.2%            | 12.0%                  | ai research and development, research and development, ai research, hybrid cloud and ai strategy |
| measure_outcome · financial outcome        | 21.2%            | 10.5%                  | ai revenue, productivity, revenue, productivity gains                                            |

*Notes:* Pooled sample covers 510 S&P 500 firms with filings and call transcripts. “Filings only” restricts the denominator to statutory SEC filings (10-K, 10-Q, DEF 14A, 8-K), illustrating the significant contribution of earnings calls to corporate disclosure of commercial tools, partnerships, and quantified metrics.

The full 14-variable disclosed behavioral profile by disclosure archetype is in [Table 25](#) below; the 6 headline variables are shown as [Table 10](#) in Chapter 4.

Table 25: Full disclosed behavioral profile by disclosure archetype (14 variables).

| Behavioral feature                                  | Vocal Substantives (n=127) | Governance-Led Disclosers (n=138) | Defensive Disclosers (n=196) | No AI (n=49) |
|-----------------------------------------------------|----------------------------|-----------------------------------|------------------------------|--------------|
| Disclosed activities per firm (Median)              | 94                         | 17                                | 8                            | 0            |
| Disclosed activities per firm (Mean)                | 169.4                      | 23.0                              | 15.3                         | 3.1          |
| Customer-facing activity share (%)                  | 51.7%                      | 22.6%                             | 24.4%                        | 22.0%        |
| Internal process activity share (%)                 | 31.6%                      | 64.9%                             | 61.5%                        | 71.9%        |
| Technology source: Proprietary / Own (%)            | 72.2%                      | 45.7%                             | 39.6%                        | 34.6%        |
| Technology source: Third-party commercial (%)       | 9.3%                       | 14.4%                             | 17.8%                        | 13.3%        |
| Technology source: Unspecified provenance (%)       | 18.5%                      | 39.9%                             | 42.6%                        | 52.1%        |
| Firms naming an own product or brand (%)            | 98.4%                      | 63.0%                             | 44.9%                        | 14.3%        |
| Firms naming an external provider or partner (%)    | 89.8%                      | 44.2%                             | 32.7%                        | 14.3%        |
| Firms reporting compute / data center capex (%)     | 89.0%                      | 48.6%                             | 40.8%                        | 8.2%         |
| Firms reporting quantified outcome metrics (%)      | 93.7%                      | 61.6%                             | 50.5%                        | 20.4%        |
| Firms naming corporate clients (%)                  | 47.2%                      | 8.7%                              | 6.1%                         | 2.0%         |
| Maximum operational stage: Scaled (%)               | 94.5%                      | 63.0%                             | 45.9%                        | 16.3%        |
| Maximum operational stage: Deployed (%)             | 5.5%                       | 31.9%                             | 39.3%                        | 18.4%        |
| Maximum operational stage: Pilot / Exploration (%)  | 0.0%                       | 1.4%                              | 5.1%                         | 12.2%        |
| Maximum operational stage: No activity reported (%) | 0.0%                       | 1.4%                              | 6.1%                         | 51.0%        |

The full prefilter specification, semantic anchors, extraction prompts, and throughput diagnostics are documented in docs/analytics/10\_pipeline.md. Corpus rows below are refreshed from the current processed data; downstream rows report the current available inference outputs.

Table 26: Complete granular inference funnel by document form.

| Stage                            | Operational Filter                | Unit         | 10-K      | 10-Q      | DEF 14A   | 8-K       | Calls   | 20-F    | Total        |
|----------------------------------|-----------------------------------|--------------|-----------|-----------|-----------|-----------|---------|---------|--------------|
| <b>A. Corpus</b>                 |                                   |              |           |           |           |           |         |         |              |
| Documents                        | Processed corpus                  | Documents    | 2,898     | 8,330     | 2,883     | 35,880    | 8,329   | 32      | 58,352       |
| Paragraphs                       | Segmentation / turns              | Instances    | 1,555,923 | 1,813,510 | 3,994,038 | 1,693,503 | 580,625 | 145,674 | 9,783,273    |
| Scorable paragraphs              | Rule: > 3 char & alnum            | Instances    | 1,432,176 | 1,659,250 | 3,527,006 | 1,595,325 | 580,256 | 140,727 | 8,934,740    |
| Unique scorable texts            | BLAKE2b content hash              | Unique texts | 861,631   | 879,602   | 2,422,715 | 421,982   | 505,417 | 106,055 | 5,123,286    |
| <b>B. Prefilter</b>              |                                   |              |           |           |           |           |         |         |              |
| Positive unique texts            | GBDT ( $\tau = 0.17, 39$ signals) | Unique texts | 11,320    | 2,944     | 6,759     | 389       | 10,596  | 151     | 31,856       |
| Share of scorable texts          | Positive rate                     | %            | 1.3%      | 0.3%      | 0.3%      | 0.1%      | 2.1%    | 0.1%    | 0.6%         |
| Positive instances               | Exact hash join                   | Instances    | 12,516    | 3,942     | 6,941     | 411       | 10,608  | 154     | 34,572       |
| <b>C. Semantic Frames</b>        |                                   |              |           |           |           |           |         |         |              |
| Classified unique texts          | Candidate set                     | Unique texts | 11,321    | 2,944     | 6,760     | 389       | 10,596  | 151     | 31,857       |
| Texts with $\geq 1$ frame        | LLM judge (Qwen 3.7)              | Unique texts | 10,559    | 2,595     | 5,554     | 234       | 8,776   | 96      | 27,521       |
| LLM acceptance rate              | Acceptance share                  | %            | 93.3%     | 88.1%     | 82.2%     | 60.2%     | 82.8%   | 63.6%   | 86.4%        |
| Unique semantic frames           | Frame extraction                  | Frames       | 16,286    | 3,419     | 7,672     | 298       | 18,106  | 119     | 45,519       |
| Document instances               | Exact hash join                   | Instances    | 17,801    | 4,603     | 7,815     | 317       | 18,129  | 122     | 48,787       |
| Documents with $\geq 1$ frame    | Document join                     | Documents    | 1,688     | 1,437     | 1,166     | 209       | 2,921   | 26      | 7,447        |
| Share of form documents          | Document share                    | %            | 58.2%     | 17.3%     | 40.4%     | 0.6%      | 35.1%   | 81.3%   | 12.8%        |
| <b>D. Disclosed Activities</b>   |                                   |              |           |           |           |           |         |         |              |
| Texts with firm conduct          | Rule: subject = firm              | Unique texts | 5,941     | 1,835     | 3,121     | 135       | 7,536   | 62      | 18,430       |
| Share of framed texts            | Behaviour share                   | %            | 56.3%     | 70.7%     | 56.2%     | 57.7%     | 85.9%   | 64.6%   | 67.0%        |
| Unique activities                | LLM pass 2 (firm ID)              | Activities   | 8,575     | 2,194     | 2,874     | 163       | 14,020  | 57      | 27,883       |
| Activities per text              | Activity density                  | Ratio        | 1.44      | 1.20      | 0.92      | 1.21      | 1.86    | 0.92    | 1.51         |
| Document instances               | Exact hash join                   | Instances    | 8,579     | 2,194     | 2,874     | 163       | 14,033  | 57      | 27,900       |
| Documents with $\geq 1$ activity | Document join                     | Documents    | 1,253     | 703       | 622       | 86        | 2,442   | 20      | 5,126        |
| Firms with $\geq 1$ activity     | Ticker join                       | Firms        | 411       | 159       | 275       | 55        | 383     | 5       | 476 (of 510) |

**Notes:** Complete granular funnel executed on `duckdb/thesis.duckdb` and `data/processed/clusters/firm_activities.parquet`, refreshed 2026-09-08. Unique-text and firm columns do not sum to total because the same text (or the same firm, across forms) can appear in more than one column.

The full distribution of disclosed activities across all six dimensions (27,883 activities, [Table 27](#) below) underlies the activity-evolution chart in Chapter 3 (“Aggregate behavioral landscape”).

Table 27: Full distribution of disclosed activities across all six dimensions.

| Action                | %     | Maturity stage | %     | Target audience       | %     |
|-----------------------|-------|----------------|-------|-----------------------|-------|
| deploy                | 62.6% | deployed       | 58.3% | customers             | 48.3% |
| invest infrastructure | 7.9%  | scaled         | 14.0% | internal process      | 25.5% |
| scale                 | 7.4%  | piloting       | 10.3% | unspecified           | 13.7% |
| pilot or explore      | 3.9%  | exploring      | 9.2%  | employees             | 8.6%  |
| develop               | 3.5%  | unspecified    | 8.2%  | partners or suppliers | 2.5%  |
| integrate             | 3.5%  |                |       | developers            | 1.3%  |
| measure outcome       | 3.4%  |                |       |                       |       |
| partner               | 3.0%  |                |       |                       |       |
| hire or train         | 2.1%  |                |       |                       |       |
| buy or license        | 1.3%  |                |       |                       |       |
| govern or control     | 1.2%  |                |       |                       |       |
| restrict              | 0.0%  |                |       |                       |       |

| AI source    | %     | Evidence strength        | %     | Business function                | %     |
|--------------|-------|--------------------------|-------|----------------------------------|-------|
| own          | 72.9% | generic                  | 53.5% | unspecified                      | 32.0% |
| unspecified  | 15.7% | named product or process | 29.2% | operations and supply chain      | 16.5% |
| third party  | 7.9%  | metric                   | 12.9% | product feature                  | 7.6%  |
| acquired     | 1.5%  | vendor                   | 4.4%  | other                            | 6.9%  |
| mixed        | 1.0%  |                          |       | it and cybersecurity             | 6.9%  |
| co developed | 0.9%  |                          |       | customer service                 | 4.6%  |
| open source  | 0.1%  |                          |       | software development             | 4.5%  |
|              |       |                          |       | marketing and sales              | 4.0%  |
|              |       |                          |       | content and media                | 2.8%  |
|              |       |                          |       | fraud and risk                   | 2.6%  |
|              |       |                          |       | data and analytics               | 2.4%  |
|              |       |                          |       | research and product development | 2.2%  |
|              |       |                          |       | hr and talent                    | 1.8%  |
|              |       |                          |       | financial services operations    | 1.8%  |
|              |       |                          |       | healthcare delivery              | 1.1%  |
|              |       |                          |       | ai compute and models            | 1.0%  |
|              |       |                          |       | finance legal and admin          | 0.8%  |
|              |       |                          |       | governance                       | 0.5%  |

*Notes:* Source: data/processed/clusters/activity\_profiles.json. “Business function” has no single dominant category comparable to customer\_service for customer-success work specifically. Customer success (retention/expansion) is not coded as a separate family from customer support in the current taxonomy.

The full named-entity roles across the corpus ([Table 28](#)) and the external technology ecosystem cited by S&P 500 corporations ([Table 29](#)) document the complete distribution of technological attribution.

Table 28: Named-entity roles across the corpus.

| Entity role                  | Instances | Share |
|------------------------------|-----------|-------|
| Own product or brand         | 13,408    | 77.5% |
| External technology vendor   | 1,195     | 6.9%  |
| Strategic commercial partner | 802       | 4.6%  |
| Enterprise customer          | 732       | 4.2%  |
| Acquired technology startup  | 556       | 3.2%  |
| External foundation model    | 474       | 2.7%  |
| Distribution channel         | 72        | 0.4%  |
| Competitor or reference      | 71        | 0.4%  |

*17,310 named entities total.*

Table 29: External AI technology ecosystem cited by S&P 500 corporations.

| External technology provider / model family | % of Firms citing | Literal named entities extracted                                     |
|---------------------------------------------|-------------------|----------------------------------------------------------------------|
| Specialized Enterprise & Data Vendors       | 42.4%             | Palantir, AMD, ServiceNow, Hugging Face, Snowflake                   |
| OpenAI / Microsoft Ecosystem                | 17.3%             | OpenAI, Microsoft, ChatGPT, Azure, GitHub Copilot                    |
| Google / Alphabet                           | 9.6%              | Google, Google Cloud, Gemini, Google AI, Google Vertex AI            |
| NVIDIA                                      | 9.2%              | NVIDIA, Nvidia, NVIDIA H100, NVIDIA AI software, NVIDIA H100s        |
| Amazon / AWS                                | 7.1%              | AWS, Amazon Web Services, Amazon, Amazon AWS, Bedrock                |
| Meta Platforms                              | 3.5%              | Meta, Llama, Llama 3, Meta’s Llama 3.1, Meta’s Llama 2               |
| Anthropic                                   | 2.4%              | Anthropic, Anthropic, PBC, Anthropic Claude, Anthropic Cloud, Claude |
| IBM                                         | 2.4%              | IBM, Watson, Notre Dame — IBM Technology Ethics Lab, watsonx, ibm    |
| Salesforce                                  | 1.4%              | Salesforce, Salesforce Platform, Einstein                            |

*Notes:* Vendor citations include entities tagged as external\_provider, external\_model, or partner. Entities appear in this table only when cited by *other* firms as an external technology source (e.g., Microsoft citing OpenAI, or Lowe’s citing Palantir and NVIDIA), excluding self-promotional brand mentions.

The sectoral distribution of frames ([Table 30](#)) and disclosed activities ([Table 31](#)) across 2-digit SIC sectors illustrates the cross-industry distribution of AI communication and concrete activity.

Table 30: Frame composition by sector (2-digit SIC, top 10 by firm count): median rates per 1,000 words.

| Sector                | Firms | Frames/1k | Promo/1k | Risk/1k | Gov/1k |
|-----------------------|-------|-----------|----------|---------|--------|
| Business services     | 66    | 0.185     | 0.024    | 0.05    | 0.021  |
| Instruments           | 42    | 0.057     | 0.002    | 0.02    | 0.006  |
| Chemicals             | 39    | 0.032     | 0.002    | 0.013   | 0.005  |
| Electric/gas/sanitary | 34    | 0.015     | 0        | 0.006   | 0.004  |

| Sector                     | Firms | Frames/1k | Promo/1k | Risk/1k | Gov/1k |
|----------------------------|-------|-----------|----------|---------|--------|
| Electronic equipment       | 29    | 0.133     | 0.015    | 0.029   | 0.009  |
| Industrial/computer equip. | 28    | 0.132     | 0.016    | 0.028   | 0.012  |
| Holding/investment offices | 27    | 0.026     | 0.001    | 0.012   | 0.004  |
| Insurance carriers         | 23    | 0.038     | 0.002    | 0.016   | 0.011  |
| Depository institutions    | 21    | 0.023     | 0.001    | 0.009   | 0.005  |
| Food products              | 18    | 0.017     | 0        | 0.008   | 0.003  |

Table 31: Disclosed activity volume and composition by sector (2-digit SIC, top 10 by firm count).

| Sector                     | Firms | Median activities/firm | Proprietary tech share |
|----------------------------|-------|------------------------|------------------------|
| Business services          | 66    | 101                    | 79.8%                  |
| Instruments                | 42    | 24                     | 67.8%                  |
| Chemicals                  | 39    | 11                     | 53.5%                  |
| Electric/gas/sanitary      | 34    | 7                      | 35.1%                  |
| Electronic equipment       | 29    | 38                     | 84.8%                  |
| Industrial/computer equip. | 28    | 54                     | 77.9%                  |
| Holding/investment offices | 27    | 9                      | 39.8%                  |
| Insurance carriers         | 23    | 22                     | 55.6%                  |
| Depository institutions    | 21    | 16                     | 52.5%                  |
| Food products              | 18    | 4                      | 38.1%                  |

## APPENDIX D: VALIDATION

The human coding protocol for the pending validation round (300 frame-labeled paragraphs / 434 frames, 300 prefilter-stratified paragraphs, 120 activity-labeled paragraphs / 267 activities, coded in the ui-validator/ tool), per-field agreement statistics once collected, confusion matrices for the promotional and temporal fields, an error taxonomy distinguishing prefilter false positives from judge misclassification, and a sensitivity analysis for collapsing or dropping fields with weak human agreement are reported here as they become available; at the time of writing, only LLM-to-LLM agreement is measured ( $\kappa$  0.81–0.87, docs/analytics/00\_funnel\_del\_corpus.md).

## APPENDIX E: STATISTICAL ROBUSTNESS

Table 32: Full M0-M3 model fit, beta.

| Outcome | R2 M0 | R2 M1 | R2 M2 | R2 M3 | Adj.R2 M0 | Adj.R2 M1 | Adj.R2 M2 | Adj.R2 M3 |
|---------|-------|-------|-------|-------|-----------|-----------|-----------|-----------|
| Beta    | 0.069 | 0.118 | 0.132 | 0.123 | -0.053    | 0.001     | 0.011     | 0.004     |

*Notes:* Beta starts with negative adjusted  $R^2$  at M0 because sector-by-year fixed effects consume many degrees of freedom relative to firms per cell, a penalty the adjustment applies even though raw  $R^2$  stays positive throughout.

Table 33: Beta: standardized coefficients, M2 content block (bold =  $p < 0.05$ ).

| Content dimension | Beta               |
|-------------------|--------------------|
| Volume            | +0.59 ( $p=0.07$ ) |
| Realized          | -0.09 ( $p=0.77$ ) |

| Content dimension | Beta                  |
|-------------------|-----------------------|
| Deployment        | -0.29 (p=0.10)        |
| Capability        | -0.04 (p=0.45)        |
| Risk              | -0.05 (p=0.55)        |
| Governance        | <b>-0.12 (p=0.03)</b> |
| Promotional       | -0.03 (p=0.78)        |
| Specificity       | +0.19 (p=0.15)        |

**AI-washing index, full specification.**  $\text{Substance}_{it} = \log(1 + n_{\text{activities},it}) \times \text{Grounding}_{it}$ , where  $\text{Grounding}_{it}$  averages, with empirical-Bayes shrinkage per component, five concreteness markers over that firm-year's disclosed activities (named function, deployed/scaled stage, named product or process, quantified outcome, named provider).  $W_{it} = \text{pctrank}(\text{Disclosure}_{it}) - \text{pctrank}(\text{Substance}_{it})$ , where  $\text{Disclosure}_{it}$  is AI frames per 1,000 words and both percentile ranks are computed within the analytical sample (firm-years with  $\text{Disclosure}_{it} > 0$ : 1,449 of 2,475 firm-years, 474 of 510 firms pooled 2021–2025). No regression, no fixed effects, no fitted residual: it is a rank-based gap between two independently extracted quantities.

Percentile rank rather than z-score standardization: with z-scores,  $\text{Disclosure}_{it}$ 's distribution has a far heavier right tail than  $\text{Substance}_{it}$ 's (maximum z of 13.7 against 3.7), which put firms with the *most* grounded activity in absolute terms (Microsoft, NVIDIA) in the washing tail whenever their disclosure happened to be an even bigger outlier than their already-large substance — Spearman between the z-score and percentile-rank versions of  $W$  is 0.988, but the specific firm-years in the top 5% differ, and only the percentile version keeps the tail composed of firms with no grounded activity behind unusually heavy disclosure.

**Substance functional form.** An additive form,  $z(\log(1 + n_{\text{activities}})) + z(\text{Grounding})$ , and a grounded-count form,  $\log(1 + n_{\text{activities}} \times \text{Grounding})$ , were both compared against the multiplicative form used here. Substance under the multiplicative and grounded-count forms correlates at Spearman = 0.993 ( $n = 1,449$ ), and the resulting  $W$  at Spearman = 0.988, with 66 of 73 top-5%-tail firm-years shared between the two (Jaccard = 0.825). The multiplicative form is used because it does not reward raw activity count over grounding quality when the two trade off directly against each other (a firm with many but poorly grounded activities does not automatically outscore one with fewer but well-grounded activities, at typical values in this sample), while the grounded-count form's arithmetic can favor the larger, less-grounded firm in exactly that comparison.

Table 34: AI-washing index validation summary (firm-year panel,  $n = 1,449$  firm-years disclosing AI).

| Check                                        | Sample      | Spearman | p      |
|----------------------------------------------|-------------|----------|--------|
| Construct consistency vs. Grounding          | n=1079      | -0.583   | <0.001 |
| Persistence ( $W_t$ vs. $W_{t+1}$ )          | n=942 pairs | 0.594    | <0.001 |
| Split-half (years split per firm)            | n=409 firms | 0.51     | <0.001 |
| Economic coherence: R&D intensity            | n=753       | -0.075   | 0.041  |
| Economic coherence: Log market cap           | n=1377      | 0.02     | 0.450  |
| Economic coherence: Beta                     | n=1373      | -0.081   | 0.003  |
| Economic coherence: Idiosyncratic volatility | n=1373      | -0.103   | <0.001 |
| Economic coherence: Price-to-sales           | n=1368      | -0.011   | 0.675  |

*Notes:* Construct consistency against Grounding is restricted to firm-years with at least one disclosed activity ( $n = 1,074$ ); Grounding is a direct component of Substance, so this is not an independent

validity check, only confirmation the score moves as its construction implies. The five economic-coherence rows test whether  $W$  tracks R&D intensity (expected negative) without also tracking size, valuation, or risk (expected flat) — the pattern that survives is a small, correctly signed R&D relationship and no material size or risk relationship.

Table 35: Firm-years in the top and bottom 5% of the AI-washing index (panel specification), with disclosure, activity, and grounding components.

| Ticker | Year | AI frames/1k words | Activities | Grounding | W    | Disclosure archetype (Chapter 4) |
|--------|------|--------------------|------------|-----------|------|----------------------------------|
| Q      | 2025 | 0.38               | 0          | 0.32      | 0.83 | Vocal Substantives               |
| SWKS   | 2025 | 0.11               | 0          | 0.32      | 0.68 | Defensive Disclosers             |
| BRK.B  | 2025 | 0.1                | 0          | 0.32      | 0.66 | Governance-Led Disclosers        |
| WRB    | 2025 | 0.09               | 0          | 0.32      | 0.64 | Defensive Disclosers             |
| ROK    | 2024 | 0.08               | 0          | 0.32      | 0.63 | Governance-Led Disclosers        |
| IFF    | 2024 | 0.08               | 0          | 0.32      | 0.63 | Defensive Disclosers             |
| LRCX   | 2025 | 0.08               | 0          | 0.32      | 0.61 | Defensive Disclosers             |
| TER    | 2025 | 0.17               | 1          | 0.15      | 0.61 | Vocal Substantives               |
| LRCX   | 2024 | 0.07               | 0          | 0.32      | 0.57 | Defensive Disclosers             |
| VRSN   | 2025 | 0.07               | 0          | 0.32      | 0.56 | Defensive Disclosers             |
| KIM    | 2024 | 0.07               | 0          | 0.32      | 0.56 | Defensive Disclosers             |
| ANSS   | 2021 | 0.06               | 0          | 0.32      | 0.56 | Vocal Substantives               |
| PM     | 2024 | 0.06               | 0          | 0.32      | 0.55 | Governance-Led Disclosers        |
| BBY    | 2024 | 0.06               | 0          | 0.32      | 0.54 | Defensive Disclosers             |
| IT     | 2024 | 0.12               | 4          | 0.1       | 0.54 | Defensive Disclosers             |
| KLAC   | 2023 | 0.06               | 0          | 0.32      | 0.53 | Defensive Disclosers             |
| CLX    | 2024 | 0.06               | 0          | 0.32      | 0.53 | Defensive Disclosers             |
| CCI    | 2024 | 0.06               | 0          | 0.32      | 0.52 | Defensive Disclosers             |
| GNRC   | 2025 | 0.18               | 2          | 0.17      | 0.5  | Defensive Disclosers             |
| WDC    | 2025 | 0.14               | 1          | 0.26      | 0.49 | Governance-Led Disclosers        |
| LMT    | 2024 | 0.1                | 3          | 0.13      | 0.49 | Governance-Led Disclosers        |
| IFF    | 2025 | 0.13               | 1          | 0.26      | 0.49 | Defensive Disclosers             |
| CMCSA  | 2025 | 0.05               | 0          | 0.32      | 0.48 | Defensive Disclosers             |
| WM     | 2025 | 0.05               | 0          | 0.32      | 0.48 | Defensive Disclosers             |
| IT     | 2025 | 0.16               | 3          | 0.17      | 0.48 | Defensive Disclosers             |
| IP     | 2025 | 0.05               | 0          | 0.32      | 0.48 | Defensive Disclosers             |
| REG    | 2025 | 0.05               | 0          | 0.32      | 0.47 | Defensive Disclosers             |
| AMP    | 2025 | 0.05               | 0          | 0.32      | 0.47 | Governance-Led Disclosers        |
| GWW    | 2025 | 0.08               | 4          | 0.1       | 0.46 | Defensive Disclosers             |
| DG     | 2025 | 0.05               | 0          | 0.32      | 0.46 | Governance-Led Disclosers        |
| CLX    | 2025 | 0.05               | 0          | 0.32      | 0.46 | Defensive Disclosers             |
| IR     | 2025 | 0.08               | 1          | 0.15      | 0.46 | Defensive Disclosers             |
| ROP    | 2025 | 0.13               | 2          | 0.17      | 0.45 | Vocal Substantives               |
| FOX    | 2025 | 0.09               | 1          | 0.26      | 0.45 | Defensive Disclosers             |
| FOX    | 2024 | 0.09               | 1          | 0.26      | 0.45 | Defensive Disclosers             |
| CTAS   | 2025 | 0.04               | 0          | 0.32      | 0.44 | Defensive Disclosers             |
| GS     | 2025 | 0.07               | 3          | 0.07      | 0.44 | Defensive Disclosers             |
| HLT    | 2025 | 0.04               | 0          | 0.32      | 0.44 | Defensive Disclosers             |
| DIS    | 2024 | 0.04               | 0          | 0.32      | 0.44 | Defensive Disclosers             |
| ROP    | 2024 | 0.1                | 1          | 0.26      | 0.43 | Vocal Substantives               |
| GM     | 2024 | 0.04               | 0          | 0.32      | 0.43 | Vocal Substantives               |
| DXCM   | 2023 | 0.04               | 0          | 0.32      | 0.43 | Vocal Substantives               |
| PM     | 2023 | 0.04               | 0          | 0.32      | 0.43 | Governance-Led Disclosers        |
| DDOG   | 2021 | 0.04               | 0          | 0.32      | 0.43 | Vocal Substantives               |

| Ticker | Year | AI frames/1k words | Activities | Grounding | W     | Disclosure archetype (Chapter 4) |
|--------|------|--------------------|------------|-----------|-------|----------------------------------|
| VMRK   | 2025 | 0.07               | 2          | 0.1       | 0.42  | Defensive Disclosers             |
| BAX    | 2025 | 0.16               | 9          | 0.11      | 0.42  | Defensive Disclosers             |
| HLT    | 2024 | 0.04               | 0          | 0.32      | 0.42  | Defensive Disclosers             |
| LRCX   | 2022 | 0.04               | 0          | 0.32      | 0.42  | Defensive Disclosers             |
| DHR    | 2024 | 0.07               | 1          | 0.15      | 0.42  | Defensive Disclosers             |
| GILD   | 2025 | 0.04               | 0          | 0.32      | 0.41  | Governance-Led Disclosers        |
| DIS    | 2025 | 0.04               | 0          | 0.32      | 0.41  | Defensive Disclosers             |
| DD     | 2024 | 0.04               | 0          | 0.32      | 0.41  | Governance-Led Disclosers        |
| DHR    | 2025 | 0.1                | 2          | 0.17      | 0.41  | Defensive Disclosers             |
| VLO    | 2025 | 0.04               | 0          | 0.32      | 0.41  | Governance-Led Disclosers        |
| SWKS   | 2024 | 0.07               | 1          | 0.26      | 0.4   | Defensive Disclosers             |
| NEE    | 2025 | 0.04               | 0          | 0.32      | 0.4   | Governance-Led Disclosers        |
| KIM    | 2025 | 0.08               | 1          | 0.26      | 0.4   | Defensive Disclosers             |
| MCD    | 2025 | 0.06               | 4          | 0.06      | 0.4   | Defensive Disclosers             |
| KLAC   | 2024 | 0.16               | 2          | 0.24      | 0.4   | Defensive Disclosers             |
| AVY    | 2025 | 0.04               | 0          | 0.32      | 0.4   | Governance-Led Disclosers        |
| A      | 2025 | 0.1                | 4          | 0.14      | 0.4   | Defensive Disclosers             |
| TDY    | 2024 | 0.1                | 2          | 0.17      | 0.4   | Defensive Disclosers             |
| DGX    | 2023 | 0.06               | 3          | 0.07      | 0.4   | Vocal Substantives               |
| SWKS   | 2023 | 0.04               | 0          | 0.32      | 0.39  | Defensive Disclosers             |
| DUK    | 2025 | 0.04               | 0          | 0.32      | 0.39  | Governance-Led Disclosers        |
| MTD    | 2025 | 0.1                | 2          | 0.17      | 0.39  | Defensive Disclosers             |
| TAP    | 2024 | 0.06               | 1          | 0.15      | 0.39  | Defensive Disclosers             |
| WHR    | 2024 | 0.06               | 1          | 0.15      | 0.39  | Defensive Disclosers             |
| GPC    | 2025 | 0.04               | 0          | 0.32      | 0.39  | Defensive Disclosers             |
| AMTM   | 2024 | 0.28               | 6          | 0.19      | 0.39  | Defensive Disclosers             |
| CDW    | 2024 | 0.08               | 1          | 0.26      | 0.38  | Defensive Disclosers             |
| NFLX   | 2024 | 0.14               | 2          | 0.24      | 0.38  | Governance-Led Disclosers        |
| WM     | 2024 | 0.04               | 0          | 0.32      | 0.38  | Defensive Disclosers             |
| LMT    | 2023 | 0.03               | 0          | 0.32      | 0.38  | Governance-Led Disclosers        |
| CPRT   | 2023 | 0.01               | 1          | 0.46      | -0.37 | Vocal Substantives               |
| VRSK   | 2023 | 0.03               | 5          | 0.45      | -0.37 | Vocal Substantives               |
| CEG    | 2022 | 0.01               | 1          | 0.46      | -0.37 | Defensive Disclosers             |
| RTX    | 2023 | 0                  | 1          | 0.35      | -0.37 | Governance-Led Disclosers        |
| AIG    | 2024 | 0.01               | 2          | 0.31      | -0.38 | Governance-Led Disclosers        |
| ICE    | 2023 | 0.04               | 8          | 0.53      | -0.38 | Vocal Substantives               |
| CVS    | 2023 | 0.03               | 5          | 0.46      | -0.38 | Governance-Led Disclosers        |
| HAL    | 2022 | 0.03               | 5          | 0.49      | -0.38 | Vocal Substantives               |
| TSLA   | 2021 | 0.01               | 4          | 0.34      | -0.38 | Vocal Substantives               |
| IRM    | 2021 | 0.01               | 1          | 0.36      | -0.38 | Vocal Substantives               |
| CVS    | 2021 | 0.01               | 3          | 0.38      | -0.38 | Governance-Led Disclosers        |
| WFC    | 2024 | 0.01               | 2          | 0.5       | -0.38 | Vocal Substantives               |
| CPRT   | 2022 | 0.01               | 1          | 0.46      | -0.38 | Vocal Substantives               |
| FMC    | 2023 | 0.01               | 1          | 0.46      | -0.38 | Defensive Disclosers             |
| CL     | 2023 | 0.02               | 4          | 0.39      | -0.38 | Governance-Led Disclosers        |
| DOW    | 2024 | 0.02               | 2          | 0.58      | -0.39 | Governance-Led Disclosers        |
| CFG    | 2022 | 0.01               | 2          | 0.38      | -0.39 | Governance-Led Disclosers        |
| BNY    | 2024 | 0.01               | 3          | 0.33      | -0.39 | Governance-Led Disclosers        |
| NEM    | 2025 | 0.01               | 3          | 0.38      | -0.39 | Governance-Led Disclosers        |
| EW     | 2021 | 0.01               | 2          | 0.5       | -0.4  | Defensive Disclosers             |
| FBHS   | 2023 | 0.01               | 2          | 0.38      | -0.4  | Governance-Led Disclosers        |
| FDX    | 2022 | 0                  | 1          | 0.36      | -0.4  | Vocal Substantives               |

| Ticker | Year | AI frames/1k words | Activities | Grounding | W     | Disclosure archetype (Chapter 4) |
|--------|------|--------------------|------------|-----------|-------|----------------------------------|
| ICE    | 2021 | 0.03               | 6          | 0.56      | -0.41 | Vocal Substantives               |
| ZS     | 2022 | 0.01               | 1          | 0.45      | -0.41 | Vocal Substantives               |
| ILMN   | 2022 | 0                  | 1          | 0.35      | -0.41 | Defensive Disclosers             |
| HOLX   | 2021 | 0.03               | 5          | 0.62      | -0.41 | Vocal Substantives               |
| CRL    | 2021 | 0.01               | 1          | 0.46      | -0.41 | Defensive Disclosers             |
| SRE    | 2022 | 0                  | 1          | 0.36      | -0.41 | Defensive Disclosers             |
| TRV    | 2022 | 0.01               | 2          | 0.31      | -0.41 | Defensive Disclosers             |
| FISV   | 2021 | 0.01               | 1          | 0.46      | -0.41 | Vocal Substantives               |
| CSCO   | 2021 | 0.02               | 5          | 0.46      | -0.41 | Vocal Substantives               |
| FLT    | 2023 | 0.01               | 3          | 0.38      | -0.42 | Governance-Led Disclosers        |
| SRE    | 2021 | 0                  | 1          | 0.36      | -0.42 | Defensive Disclosers             |
| CTXS   | 2021 | 0.01               | 2          | 0.51      | -0.42 | Vocal Substantives               |
| ALK    | 2023 | 0.01               | 2          | 0.52      | -0.42 | Governance-Led Disclosers        |
| WELL   | 2022 | 0                  | 1          | 0.36      | -0.42 | Defensive Disclosers             |
| ZS     | 2021 | 0.01               | 1          | 0.45      | -0.42 | Vocal Substantives               |
| WELL   | 2023 | 0                  | 1          | 0.36      | -0.42 | Defensive Disclosers             |
| EBAY   | 2021 | 0.01               | 2          | 0.31      | -0.43 | Vocal Substantives               |
| FCX    | 2021 | 0                  | 1          | 0.36      | -0.43 | Governance-Led Disclosers        |
| CRL    | 2024 | 0.01               | 1          | 0.46      | -0.43 | Defensive Disclosers             |
| SYF    | 2021 | 0                  | 1          | 0.36      | -0.43 | Governance-Led Disclosers        |
| CFG    | 2023 | 0                  | 1          | 0.36      | -0.43 | Governance-Led Disclosers        |
| CFG    | 2021 | 0                  | 1          | 0.36      | -0.44 | Governance-Led Disclosers        |
| CAH    | 2024 | 0.02               | 5          | 0.49      | -0.44 | Governance-Led Disclosers        |
| AES    | 2023 | 0                  | 1          | 0.37      | -0.44 | Vocal Substantives               |
| BIIB   | 2022 | 0.01               | 2          | 0.45      | -0.44 | Governance-Led Disclosers        |
| C      | 2021 | 0                  | 1          | 0.36      | -0.44 | Governance-Led Disclosers        |
| CRL    | 2023 | 0.01               | 1          | 0.46      | -0.44 | Defensive Disclosers             |
| UHS    | 2024 | 0                  | 1          | 0.37      | -0.44 | Governance-Led Disclosers        |
| MDT    | 2021 | 0.02               | 6          | 0.39      | -0.45 | Vocal Substantives               |
| CARR   | 2022 | 0.01               | 2          | 0.59      | -0.45 | Vocal Substantives               |
| CVS    | 2022 | 0.01               | 3          | 0.43      | -0.46 | Governance-Led Disclosers        |
| ALK    | 2021 | 0.01               | 1          | 0.46      | -0.46 | Governance-Led Disclosers        |
| CE     | 2024 | 0.01               | 2          | 0.45      | -0.47 | Governance-Led Disclosers        |
| MMM    | 2022 | 0.01               | 2          | 0.51      | -0.47 | Governance-Led Disclosers        |
| WYNN   | 2021 | 0.01               | 1          | 0.58      | -0.48 | Governance-Led Disclosers        |
| RF     | 2025 | 0.03               | 18         | 0.56      | -0.48 | Governance-Led Disclosers        |
| ICE    | 2022 | 0.02               | 7          | 0.54      | -0.48 | Vocal Substantives               |
| HON    | 2023 | 0                  | 1          | 0.46      | -0.49 | Vocal Substantives               |
| AWK    | 2024 | 0.01               | 6          | 0.19      | -0.5  | Vocal Substantives               |
| TFC    | 2022 | 0                  | 1          | 0.46      | -0.5  | Governance-Led Disclosers        |
| GE     | 2023 | 0                  | 1          | 0.46      | -0.51 | Defensive Disclosers             |
| CARR   | 2023 | 0.01               | 3          | 0.48      | -0.51 | Vocal Substantives               |
| GNRC   | 2021 | 0.02               | 5          | 0.52      | -0.51 | Defensive Disclosers             |
| SYF    | 2022 | 0                  | 1          | 0.48      | -0.52 | Governance-Led Disclosers        |
| KHC    | 2022 | 0.01               | 2          | 0.38      | -0.52 | Governance-Led Disclosers        |
| EIX    | 2025 | 0.01               | 3          | 0.38      | -0.52 | Governance-Led Disclosers        |
| CFG    | 2024 | 0                  | 2          | 0.38      | -0.55 | Governance-Led Disclosers        |
| GE     | 2021 | 0.02               | 7          | 0.54      | -0.59 | Defensive Disclosers             |
| O      | 2021 | 0                  | 2          | 0.44      | -0.6  | Governance-Led Disclosers        |
| O      | 2023 | 0.01               | 4          | 0.46      | -0.61 | Governance-Led Disclosers        |
| AEP    | 2025 | 0                  | 2          | 0.45      | -0.61 | Governance-Led Disclosers        |
| CI     | 2021 | 0.01               | 4          | 0.46      | -0.64 | Governance-Led Disclosers        |

*Notes:* Positive  $W$  is the washing tail (disclosure ranks higher than substance); negative  $W$  is the quiet-substance tail (substance ranks higher than disclosure). Both tails together cover 146 of the 1,449 scoreable firm-years (top and bottom 5%).

**Alternative firm-year aggregation.** Results are unchanged when firm-year intensity is constructed as an aggregate-numerator-over-aggregate-denominator ratio rather than the median document rate: the two rollups are numerically identical across all seven content dimensions and 2,475 firm-years (max absolute difference below  $1e-9$ ), because a firm-year's AI content comes from a single dominant filing often enough that the rollup rule carries no weight in practice.

## APPENDIX F: VARIABLE CONSTRUCTION

Table 36: XBRL concept coverage.

| Financial concept | Coverage of the firm-year panel               |
|-------------------|-----------------------------------------------|
| Revenue           | 99%                                           |
| Net margin        | 99%                                           |
| Capex             | 91%                                           |
| SG&A              | 81%                                           |
| R&D expense       | 46% (1,318 of 2,868 firm-years; outcome only) |

Fiscal-period alignment rules (period-end date matched to the nearest prior filing date; the [340, 380]-day validity window for year-over-year growth), the market-model beta and idiosyncratic-volatility construction (252 trading days pre-filing, one-factor market model), and the ROIC-minus-WACC construction (geometric equity risk premium of 6.48%, not arithmetic) are documented here.

### 8.0.3 R&D COVERAGE AND INTERPRETATION

R&D expense is constructed from four comparable current-expense concepts: `us-gaap:ResearchAndDevelopmentExpense`, `us-gaap:ResearchAndDevelopmentExpenseExcludingAcquiredInProcessCost`, `us-gaap:ResearchAndDevelopmentExpenseSoftwareExcludingAcquiredInProcessCost`, and `ifrs-full:ResearchAndDevelopmentExpense`. The latter two recover software reporters and IFRS filers that the narrower US-GAAP mapping missed; they add 34 firm-years to the panel. The resulting measure is available for 1,318 of 2,868 firm-years (46.0%).

This is not evidence that the remaining firms spend zero on R&D. In a SIC-based diagnostic grouping, separately reported R&D is common among manufacturing and software firms (81.4% coverage) and almost absent in financial services and real estate (1.1%). The missingness is therefore primarily an economic and disclosure-selection feature of the accounting data, rather than a generic failure of the XBRL collection process. The raw stored facts contain no company-specific extension namespace, so this pipeline cannot recover issuer-defined R&D extensions from that input.

For that reason, R&D-to-sales is not imputed and is excluded from the common set of financial controls. When it is an outcome, estimates are interpreted for the non-random subsample of firms that separately disclose comparable current R&D; each specification reports its effective sample size. Tags for tax credits, capitalized development, acquired in-process R&D, and industry-specific exploration costs are deliberately not pooled into this measure because they do not represent the same current R&D expense.

### 8.0.4 MARKET BETA AND ESTIMATION-WINDOW DIAGNOSTICS

For each firm-year, market beta is estimated from a one-factor market model over the 252 trading days strictly before the actual filing date of the 10-K. Let  $R_{i,t}$  denote the daily return on the firm's adjusted close,  $R_{f,t}$  the daily risk-free rate, and  $(\text{MKT} - \text{RF})_t$  the daily market excess return from the Ken French Fama–French three-factor file (ff3\_daily.parquet). The estimated regression is

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i(\text{MKT} - \text{RF})_t + \varepsilon_{i,t}.$$

build\_market\_factors.py identifies the first trading day on or after the filing date and takes the preceding 252 price observations. It then retains only dates for which the firm return,  $(\text{MKT} - \text{RF})_t$ , and  $R_{f,t}$  are all available, and estimates the intercept and slope by OLS. The code requires at least 120 valid daily observations (BETA\_MIN\_OBS = 120); otherwise beta is recorded as missing. The idiosyncratic-volatility measure uses the standard deviation of residuals from the same fitted model, annualized by  $\sqrt{252}$ .

The resulting firm\_year\_market\_factors.parquet panel contains 2,762 firm-years for 477 tickers in 2021–2026. Beta is available for 2,739 observations (99.2%). The estimation-window diagnostic has a mean of 249.8 return days and a median of 252; 98.0% of estimates use at least 240 trading days. The remaining 23 firm-years have fewer than 120 valid observations and are retained with beta missing rather than receiving a short-window estimate. Among valid estimates, beta has mean 0.91, median 0.88, standard deviation 0.47, interquartile range 0.59–1.18, and range  $[-0.31, 2.96]$ .

Daily prices are collected from 2020-01-01, so early-2021 filings have a prior-year trading history available for the 252-day window. The sampling frame is built around S&P 500 membership frozen at 2021-12-31, supplemented by the manual core; 29 of the 517 firms in the current universe are explicitly marked delisted. Price collection first uses Yahoo Finance and then falls back to Stooq and Tiingo when a delisted ticker is unavailable. Freezing membership and retaining these firms reduces survivorship bias relative to a current-constituent sample, although historical price coverage still depends on the availability and adjustment quality of those fallback sources.

## APPENDIX G: SEC / DEEPSEEK SECONDARY SPECIFICATIONS

Secondary SEC event-study specifications (alternative window widths, weighted variants, cells excluding mixed pre/post filings) and alternative event-window specifications for both shocks are reported here in full, consistent with the one-sentence summary in Chapter 5: no discontinuity separable from prior trends was identified for DeepSeek, and the analysis is not extended to outcome variables (competitive response, generative-AI capex) that a valid identification would be needed to interpret.

Table 37: DeepSeek event study (2025Q1 cutoff, 5,289 firm-quarters, 484 firms): pre-trend test by outcome.

| Outcome              | Pre-trend test p-value | Fails at 5%? |
|----------------------|------------------------|--------------|
| Promotional          | 0.715                  | No           |
| Specificity          | 0.638                  | No           |
| Risk                 | <0.001                 | Yes          |
| Hypothetical framing | 0.089                  | No           |
| Total AI frames      | 0.007                  | Yes          |

Notes: Risk disclosure and total AI frames fail the pre-trend test ( $p < 0.01$ ). For promotional tone, specificity, and hypothetical framing, pre-trends are flat ( $p > 0.05$ ), but post-event coefficients are

statistically indistinguishable from zero ( $p > 0.10$ ). Consequently, no causal discontinuity around the January 2025 shock is identified.